ALL-TIME APPROXIMATION OF HERMITIAN–YANG–MILLS FLOW ON DEGENERATING CURVES

BERND JOHANNES WUEBBEN

Abstract. We prove an all-time approximation of Hermitian–Yang–Mills flow by a finite matrix gradient flow on a degenerating family of compact genus-three curves. Four independent node parameters and two component-holonomy parameters determine flat unitary bundles passing from stable to strictly polystable. With fixed infinitesimal holonomy generators and node identifications, the spatially integrated squared error is bounded, uniformly in time, by the initial integrated trace discrepancy plus an error vanishing as all six parameters tend independently to zero. Actual initial metrics are uniformly bounded and finite initial data lie in a small ball; no lower bound on the weakest decay rate is required. An open spatial initial class enters after preparation negligible on every finite relaxation scale. The proof writes the first-order component-frame error as a global divergence using its zero angular mean, and combines HYM energy with time-integrated bounds from the finite gradient flow. We also prove relative asymptotics for every positive small eigenvalue of the endomorphism connection Laplacian, with exact correspondence of kernels and varying node identifications. One stable family has relaxation scales of orders h^{-1}, h^{-2} and h^{-4} as $h \to 0$, all covered by the nonlinear theorem.

1. Introduction

Hermitian–Yang–Mills flow on a degenerating curve can have several relaxation scales. Long annuli weaken the interaction between normalization components, while a component connection approaching a reducible connection introduces further small endomorphism modes. An approximation on a fixed multiple of the shortest annular length need not reach the time at which the weakest mode relaxes. The problem is to compare actual trajectories with a finite matrix evolution on the entire positive time axis, allowing these two sources of small eigenvalues to vary independently.

We consider a compact genus-three plumbing family with two elliptic and two rational normalization components. Four independent node parameters control the annular lengths. Two independent real parameters control flat unitary connections on the elliptic components. Fixed unitary identifications at the nodes give a flat bundle on the entire family; its component and cycle holonomies can be noncommuting. The finite configuration space is a product of spaces of positive Hermitian matrices of determinant one. Its potential is the sum of component commutator potentials and squared distances across the nodes, with their geometrically determined weights.

The all-time comparison uses two different estimates for the error of the reconstructed metric. The component and radial errors have an integrable bound in terms of the finite forces. The first-order change from component frames to parallel node frames has only the

Date: November 11, 2026
This version: November 11, 2026.
2020 *Mathematics Subject Classification.* 53C07 (primary); 14D06, 58J35, 58J50 (secondary).
Key words and phrases. Hermitian–Yang–Mills flow, nodal degeneration, flat bundle, small eigenvalue, matrix gradient flow, gluing.

first angular Fourier modes. A Neumann problem on each incidence collar writes this error as a divergence whose space-time energy is small. The actual HYM energy identity then controls its contribution to the trace comparison. This argument does not divide by the smallest positive eigenvalue.

Throughout the all-time theorem, the component matrices and node identifications are fixed. Only their nonnegative weights vary, and the initial finite metrics lie in a specified small ball. Zero component weights and nonzero common kernels are permitted. The approximation is an absolute integrated squared-distance estimate; it does not compare errors relative to a trajectory that is tending to zero.

1.1. **The main theorem.** Let C_τ be the smooth fibre of the four-node family constructed in Section 2. Its normalization components have area one. On the elliptic components the reference connections are $d - i\alpha_i T_i \vartheta_i$, where T_i are fixed trace-free Hermitian matrices and ϑ_i fixed nonzero real harmonic one-forms. The rational component connections are trivial. The parameters α_i are real and independent of the four complex node parameters $\tau_e \neq 0$. Write h_0 for the global parallel bundle metric and set

$$L_e = \log(1/|\tau_e|), \quad L_* = \min_e L_e, \quad \epsilon_e = L_e^{-1}, \quad \alpha_* = \max_i |\alpha_i|, \quad \kappa_i = \int_{E_i} |\vartheta_i|^2.$$

The joint limit is $\alpha_* \to 0$ and $L_* \to \infty$. All parameters are held constant during each flow.

Let $\mathcal{P}_r$ be the positive determinant-one Hermitian matrices, with metric $\langle V, W\rangle_H = \mathrm{tr}(H^{-1}VH^{-1}W)$ and distance d. Equip $\mathcal{P}_r^4$ with the product metric g_A weighted by the component areas $A_i = 1$. The finite evolution is the negative gradient flow of

$$\Phi(H) = \sum_{i=+,-} \alpha_i^2 \kappa_i \{\mathrm{tr}(T_i H_i^{-1} T_i H_i) - \mathrm{tr}(T_i^2)\} + \pi \sum_{e=ij} \epsilon_e d(H_i, U_{ij} H_j U_{ij}^*)^2. \tag{1}$$

Its matrix equation is printed in (12). The determinant-one actual HYM equation is $h^{-1}\partial_s h = -2i\Lambda F_h$ in physical time s. For two metrics define $D_{\mathrm{tr}}(h,k) = \mathrm{tr}(h^{-1}k + k^{-1}h - 2I)$.

Theorem 1.1. *Fix the rank, component metrics, matrices T_i, harmonic forms ϑ_i and unitary node identifications U_e. There is $R_* > 0$ and an explicit smooth reconstruction $H \mapsto \widehat{K}_{\tau,\alpha}(H)$ such that the following holds for sufficiently small joint parameters. Let $H(s)$ be the finite gradient flow from distance at most R_* from (I,I,I,I). Let $h(s)$ be actual HYM flow on the same smooth fibre, from any smooth metric with the parallel determinant and fixed bounds $ch_0 \le h(0) \le C_0 h_0$. Then*

$$\sup_{s\ge 0} \int_{C_\tau} d(h(s), \widehat{K}_{\tau,\alpha}(H(s)))^2 \, dA_\tau \le \eta + C\left(\alpha_* + \frac{(1+\log L_*)^2}{L_*}\right), \tag{2}$$

where $\eta = \int_{C_\tau} D_{\mathrm{tr}}(h(0), \widehat{K}_{\tau,\alpha}(H(0)))\, dA_\tau$. Constants are independent of all $\alpha_i^2 L_e$, all L_e/L_f, and time. Component weights may vanish and the common space of parallel trace-free endomorphisms may be nonzero. On each fixed compact normalization region the reconstruction differs from $H_i(s)$ by $O(L_*^{-1})$.

The proof is Theorem 6.2. The exponentially small geometric-transfer term in (60) is absorbed in the error displayed here. Uniform approximation requires $\eta \to 0$ over the chosen initial family. Theorem 7.2 removes the need to prescribe the finite initial tuple. It gives a nonempty open class of spatially varying smooth metrics, defined by the component extension norm (79), which enters the comparison at a common time $s_* \to \infty$ satisfying $(\alpha_*^2 + L_*^{-1})s_* \to 0$. The selected tuple depends on the initial component profiles. Smooth middle-neck data need only obey the fixed pointwise bound; their derivatives need not be bounded uniformly.

Theorem 8.1 identifies every small eigenvalue of the actual endomorphism connection Laplacian. Its finite energy is the sum of $\alpha_i^2\kappa_i\|[T_i,X_i]\|^2$ and $(2\pi/L_e)\|X_i-\mathrm{Ad}(U_e)X_j\|^2$, with mass $\sum_i A_i\|X_i\|^2$. The comparison is relative for every positive weak eigenvalue, identifies the exact kernel, and leaves a uniform complementary gap. In contrast to the all-time nonlinear constants, this spectral result is uniform also when the transports vary through reducible tuples.

The same family contains a stable rank-two example with three node weights h, a fourth node weight h^2, and component weight $\alpha_+^2\kappa_+ = h^4$. Its twelve small eigenvalues comprise nine order-h modes, two asymptotic to $\frac{1}{2}h^2$, and one asymptotic to h^4. Corollary 8.2 proves both these spectral claims and actual nonlinear approximation through h^{-4} and all later times. The latter conclusion uses the all-time residual argument, independently of the spectral asymptotics.

1.2. **The analytic mechanism.** The reconstruction uses scalar component Green functions, exactly flat geodesic node profiles and relative-logarithm interpolation on collars of radius $\rho = L_*^{-1}$. The frame change on an elliptic incidence is $e^{i\alpha_i T_i F_{ie}}$ with F_{ie} a real linear harmonic function in the branch coordinate. Estimating its cutoff error in absolute L^1 can produce an integral of size $1/|\alpha_i|$; the explicit calculation (23) explains this difficulty.

Instead, the term linear in α_i in the full Chern expression has only angular Fourier indices $1,-1$ after the conformal area factor is removed. Its entrywise zero mean allows a collar Neumann solve with zero normal trace at both boundaries. Extension by zero gives a global divergence without a boundary measure. Thus the residual is D^*J+B, with J small in space-time L^2 and B small in space-time L^1. The trace comparison pairs J with derivatives of the actual and reconstructed metrics; both have uniformly bounded total energy.

For the remainder B, a fixed-matrix weighted normal equation estimate controls the separate finite forces by the gradient on a small ball. Ohta's bounded-image rectifiability theorem for self-contracted curves [Oht20] then gives their uniform total integral. The fixed-constraint restriction enters at this step. The matrix counterexample in Section 3 shows why a uniform lower bound for the full unweighted matrix alone does not suffice.

1.3. **Relation to earlier work.** Burger's theorem gives relative scalar graph asymptotics on hyperbolic surfaces with unrestricted ratios among the short geodesic lengths [Bur90]. Ji–Wentworth prove spectral convergence for conic degenerations [JW92, JW97]; their correction replaces a false isoperimetric lemma, which is not used here. Daskalopoulos–Wentworth establish connection-valued spectral convergence and uniform Yang–Mills estimates away from accidental reducibility [DW00, Theorem 4.13, Corollary 4.14 and Proposition 5.8]. Our normalization components carry reducible holonomy, so their uniform-gap argument does not supply the additional slow dynamics considered here.

Haiden–Katzarkov–Kontsevich–Pandit develop finite-dimensional asymptotic descriptions of HYM flow on a fixed base [HKKP18]. Their Proposition 4.5 also compares exact and approximate metrics under an integrable pointwise bound on the forcing. The divergence estimate here uses spatially integrated energy to treat a first-order collar term not controlled uniformly under degeneration by that pointwise criterion. Lam's fixed-base asymptotic theorem for semistable bundles assumes a locally free graded bundle, with additional rationality hypotheses in its detailed reduction [Lam26]. Applying these results fibre by fibre does not give constants uniform under pinching.

Previous work on nodal curves gives actual noncommuting HYM approximation on bounded slow-time intervals [Wue26a] and treats nontrivial gluing, spatial entry and all-time

comparison under a positive normalized rate [Wue26b]. The present result removes that rate requirement for fixed constraints on a small basin and couples independent node and component-holonomy scales in one compact family. The new all-time step is the global divergence decomposition of the full Chern residual; the finite graph operator or the scalar multiscale principle alone is not that step.

2. A compact family with independent parameters

We first construct the compact family and fix the gauges used in the analysis. The construction is classical pointed plumbing [HK14, §8.3], with a particular choice of component metrics and flat connections. The auxiliary markings will be forgotten without contracting any component.

2.1. **The curve and its metrics.** Let E be the smooth projective completion of

$$y^2 = (x-1)(x-2)(x-3),$$

and let p, q be the two points over $x = 0$. Take copies E_+, E_- of E and copies R_p, R_q of $\mathbb{P}^1$. Join the two points named p to $0, \infty$ on R_p, and join the two points named q to $0, \infty$ on R_q. The resulting nodal curve C_0 has dual graph $K_{2,2}$. Its arithmetic genus is $1+1+1 = 3$. The points $1 \in R_p, R_q$ make it a stable two-pointed curve.

Choose disjoint coordinate disks at all eight branch points, using flat coordinates on the elliptic curves and unitary Möbius coordinates on the rational curves. With the components and these coordinates fixed, pointed plumbing gives a proper flat family

$$\pi : \mathcal{C} \longrightarrow B = \Delta_\delta^4, \qquad u_e v_e = \tau_e \quad (e \in E(K_{2,2})). \tag{3}$$

Here $E(K_{2,2})$ denotes the edge set, not the elliptic curve. All four τ_e are independent. For clarity, choose fixed radii $0 < a < b_0$ in the coordinate disks and $\delta < a^2$. The product component exteriors are glued to the patches $|u_e|, |v_e| < b_0$ by

$$(u_e, \tau) \longmapsto (u_e, \tau_e/u_e, \tau_{f \neq e}),$$

and the analogous map on the other branch. The two incidence overlaps of one node are disjoint, since both $|u_e| > a$ and $|v_e| > a$ would imply $|\tau_e| > a^2$. Different node patches are disjoint. There are no triple incidence overlaps. Locally the total space has coordinates $(u_e, v_e, \tau_{f \neq e})$ and is smooth. Every fibre with all $\tau_e \neq 0$ is a connected smooth compact curve of genus three. Forgetting the two sections leaves the same proper family. In particular, the rational bridges in C_0 remain present. Independence of these family parameters makes no assertion about injectivity into unmarked moduli.

Give E_+, E_- their flat metrics of area one and R_p, R_q their Fubini–Study metrics of area one. Write ω_i^{area} for these Kähler forms and $A_i = 1$ for their areas.

Lemma 2.1. *After shrinking B, the total space $\mathcal{C}$ has a smooth Kähler form whose restriction to each central normalization component is ω_i^{area}. Each smooth fibre has area four. On a sufficiently small node patch its fibre metric is the restriction of $\omega_i^{\mathrm{area}}(u_e) + \omega_j^{\mathrm{area}}(v_e)$.*

Proof. Choose on each component a degree-one holomorphic line bundle L_i with a Hermitian metric of Chern form ω_i^{area}. On an elliptic component, start with any metric on a degree-one line bundle and solve the scalar Poisson equation for the difference of the two curvature forms. That difference has integral zero. Fourier series on the flat torus give a smooth solution. On $\mathbb{P}^1$, take $\mathcal{O}(1)$. We use the convention in which a degree-one Chern form integrates to one.

In a holomorphic frame near a branch point write the squared norm as $e^{-\psi_i(u)}$, with $\psi_i(0) = 0$. On an elliptic branch a holomorphic frame change removes the harmonic part of

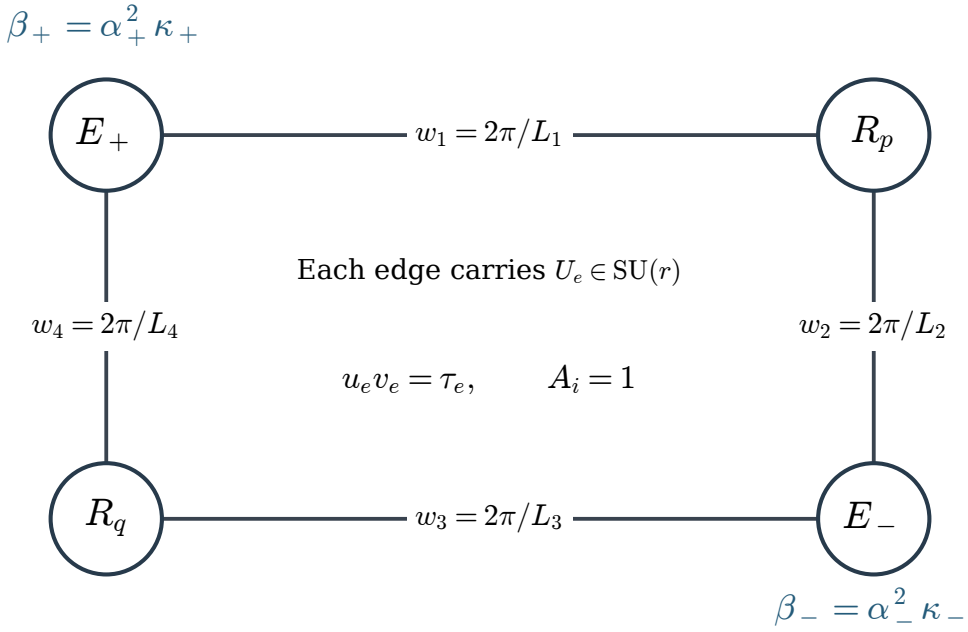


FIGURE 1. The four normalization components and their independent parameters. The edge coefficients are $w_e = 2\pi/L_e$ and the two component coefficients are $\beta_i = \alpha_i^2 \kappa_i$. The rational bridges are retained in the compact family.

the potential, leaving a positive multiple of $|u|^2$. On a rational branch the chosen coordinate gives a positive multiple of $\log(1+|u|^2)$. In particular, the first derivative of every ψ_i vanishes at zero.

Glue the product line bundles on the component exteriors to a trivial line on each node patch by these frames. The absence of triple incidence overlaps verifies the cocycle condition. On a smaller node patch set the squared norm of its frame equal to

$$\exp\{-\psi_i(u_e) - \psi_j(v_e)\}.$$

On the u_e overlap its ratio to the component metric is $\exp\{-\psi_j(\tau_e/u_e)\}$. Interpolate the logarithm of this ratio on a fixed incidence collar, leaving the product metric unchanged outside it. The resulting smooth metric on the line bundle has Chern form Ω. Along a central branch the correction and all its tangential derivatives vanish, so its restriction is exactly the prescribed component form. On the smaller node patches its curvature is the sum of the two branch forms.

Let ω_B be the Euclidean form on B. Near a node, $\Omega + C\pi^*\omega_B$ is positive for every $C > 0$: the branch forms control the u_e, v_e directions and the base form controls the remaining three coordinates. On the compact remainder of the central fibre, π is a submersion and Ω is strictly positive on the vertical tangent. In finitely many submersion charts, a Schur-complement estimate makes $\Omega + C\pi^*\omega_B$ positive for one sufficiently large C. Properness and a smaller B make these neighborhoods cover every fibre. The form is closed and of type

$(1,1)$, hence Kähler. Its fibre restrictions do not depend on the added base form. The degree of the line bundle on every smooth fibre is $\sum_i \deg L_i = 4$. □

Write dA_τ for this fibre area form and define

$$L_e = \log(1/|\tau_e|), \quad L_* = \min_e L_e, \quad \epsilon_e = L_e^{-1}, \quad \epsilon_* = L_*^{-1}, \quad b = \max_e |\tau_e| = e^{-L_*}. \tag{4}$$

All limits below have $L_* \to \infty$.

Lemma 2.2. *Choose a radius ρ with $b < \rho^2$. The middle annulus of node e, given by $|\tau_e|/\rho < |u_e| < \rho$, has area at most $C\rho^2$, independently of its modulus. Its scalar Dirichlet capacity is*

$$\frac{2\pi}{L_e + 2\log\rho}. \tag{5}$$

The punctured component $|u_e| \geq \rho$ has a natural holomorphic identification with its image in C_τ. For every fixed derivative order m, its fibre metric differs from the component metric by at most $C_m b^2 \rho^{-N_m}$ in that order, for some finite N_m.

Proof. The two branch terms in Lemma 2.1 each integrate over a disk of radius ρ, proving the area bound. In complex dimension one the Dirichlet integral is conformally invariant. The radial harmonic function between the two boundary circles is affine in $\log|u_e|$, and direct integration of its squared derivative gives (5). In the branch identification, $v_e = \tau_e/u_e$. Differentiating this identity in the fixed branch potentials gives the stated polynomial losses in ρ^{-1} and the factor $|\tau_e|^2$. On the fixed interpolation collars the denominators are bounded, and outside them the metric agrees with the product metric. No inverse power of an individual τ_e occurs. In particular $\rho = L_*^{-1}$ makes every required error tend to zero. □

2.2. **The flat bundle.** Fix $r \geq 2$, trace-free Hermitian matrices T_+, T_-, and nonzero real harmonic one-forms ϑ_+, ϑ_- on the two flat elliptic curves. For two small real numbers α_+, α_- set

$$D_i = d - i\alpha_i T_i \vartheta_i \quad (i = +,-), \qquad D_i = d \quad (i = p,q), \qquad \kappa_i = \int_{E_i} |\vartheta_i|^2 \, dA_i. \tag{6}$$

The elliptic connections are expressed in global smooth unitary frames. Their curvature and determinant connection vanish. Near an elliptic branch point the primitive of ϑ_i vanishing at the point is

$$F_{ie}(u) = c_{ie}^{\mathrm{harm}} u + \overline{c_{ie}^{\mathrm{harm}} u}. \tag{7}$$

This identity is exact: the coordinates are flat and the harmonic one-form is constant. The change of frame $G_{ie} = \exp(i\alpha_i T_i F_{ie})$ is unitary and gives a parallel frame. On rational branches put $G_{ie} = I$.

Fix unitary matrices $U_{ij} \in \mathrm{SU}(r)$ on oriented edges, with $U_{ji} = U_{ij}^{-1}$. We use the convention that they carry the j-end data to the i-end frame. Glue a trivial flat bundle on each node patch to the component local systems using constant identifications in these parallel frames, with relative matrix U_{ij}.

Proposition 2.3. *For every fixed $\alpha = (\alpha_+, \alpha_-)$ these identifications define a flat unitary connection D and parallel determinant on a holomorphic vector bundle over the whole smooth total space $\mathcal{C}$. They depend smoothly on α. The four complex parameters τ_e and the two real parameters α_i vary independently. The parallel Hermitian metric h_0 restricts to an HYM metric of slope zero on every smooth fibre.*

Proof. Pull each component connection back to its product exterior, taking the full exterior derivative there. It is flat. On every incidence overlap, the designated frames are parallel, and the constant node identifications are isomorphisms of flat connections. There are no triple incidence overlaps. The connections and metrics therefore glue. Flatness implies $(D^{0,1})^2 = 0$, so local parallel frames are holomorphic and define the asserted locally free bundle on $\mathcal{C}$. All transition maps have determinant one. This construction uses no relation between α and τ and depends smoothly on the real parameters. We assert holomorphic dependence on τ for each fixed α, not a flat connection in the additional real parameter directions. The fibrewise Chern curvature of h_0 vanishes. □

2.3. **Stability and parallel endomorphisms.** Shrinking the allowed α interval once, the parallel trace-free Hermitian endomorphisms of D are exactly the tuples satisfying

$$X_i = \mathrm{Ad}(U_{ij})X_j \quad (e = ij), \qquad \alpha_i[T_i, X_i] = 0 \quad (i = +, -). \tag{8}$$

Indeed, choose a nonzero period of ϑ_i. For sufficiently small nonzero α_i, distinct eigenvalues of T_i remain distinct after exponentiation by that period. Its holonomy commutant is thus the commutant of T_i. A commuting endomorphism is constant in the harmonic gauge and in each node-normalized parallel frame. Node compatibility is precisely the first equation of (8). When $\alpha_i = 0$ the second equation is vacuous.

The unitary-flat correspondence [NS65] makes every smooth-fibre bundle polystable of degree zero. It is stable exactly when (8) has no nonzero solution. Equivalently, the degree formula for a holomorphic subbundle expresses its degree as the negative squared norm of its second fundamental form. Equality forces a parallel orthogonal splitting. Conversely a nonzero trace-free Hermitian commutant supplies such a splitting by its eigenspaces.

For an explicit transition in rank two, let

$$T_+ = \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \qquad \alpha_- = 0, \qquad U = \frac{I + i\sigma_x}{\sqrt{2}}, \qquad \sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}. \tag{9}$$

Take the identity on a spanning tree and U on the remaining edge. Tree compatibility makes all X_i equal. For $\alpha_+ \neq 0$, the component commutant is the σ_z axis, whereas the cycle edge fixes the σ_x axis. Their intersection is zero. At $\alpha_+ = 0$ the latter axis survives. Thus the same compact family is stable for sufficiently small $\alpha_+ \neq 0$ and strictly polystable at $\alpha_+ = 0$.

The component and cycle operations in this example do not commute. To interpret cycle holonomy, fix basepoints and paths to the nodes. The resulting based holonomy includes component path factors $\exp(i\alpha_i T_i \int \vartheta_i)$ as well as the U_{ij}. A product of node matrices alone is not generally the based holonomy. The family changes holomorphic stability by changing the bundle, not by changing the polarization of a curve. Its underlying smooth $\mathrm{SU}(r)$ bundle is topologically trivial; no unstable locus is claimed.

3. The finite evolution and its total force

Let $\mathcal{P}_r$ be the space of positive Hermitian $r \times r$ matrices of determinant one. Its affine-invariant metric is

$$\langle V, W\rangle_H = \mathrm{tr}(H^{-1}VH^{-1}W), \qquad d(H, K) = \|\log(H^{-1/2}KH^{-1/2})\|_{\mathrm{HS}}.$$

It is a finite-dimensional Hadamard manifold. On $\mathcal{P}_r^4$ use the product metric $g_A = \sum_i A_i g_i$ and write $o = (I, I, I, I)$. The masses are $A_i = 1$ in our family; retaining them in the formulas indicates the origin of each coefficient. All matrix norms below are Hilbert–Schmidt norms unless stated otherwise.

For a Hermitian matrix T set

$$f_T(H) = \operatorname{tr}(TH^{-1}TH) - \operatorname{tr}(T^2). \tag{10}$$

On the family of Section 2 define

$$\Phi(H) = \sum_{i=+,-} \alpha_i^2 \kappa_i f_{T_i}(H_i) + \pi \sum_{e=ij} \epsilon_e d(H_i, U_{ij} H_j U_{ij}^*)^2. \tag{11}$$

The negative gradient equation is

$$H_i^{-1}\dot{H}_i = -\frac{\alpha_i^2\kappa_i}{A_i}[T_i, H_i^{-1}T_iH_i] + \frac{2\pi}{A_i}\sum_{e=ij}\epsilon_e B_{ij}, \qquad B_{ij} = \log(H_i^{-1}U_{ij}H_jU_{ij}^*). \tag{12}$$

The component term is omitted at rational vertices. The logarithm in B_{ij} is the principal matrix logarithm; it is H_i-self-adjoint and trace-free. The edge summation uses the orientation from the vertex under consideration. To check the component term, vary H by HX, with X H-self-adjoint and trace-free. Differentiating (10) gives

$$df_T(H)[HX] = \operatorname{tr}\big([T, H^{-1}TH]X\big).$$

The gradient of half the squared distance to K is $-H\log(H^{-1}K)$, giving the edge term. The later reconstruction derives the same coefficients from the Chern equation on the compact fibre.

The following finite estimate is useful because it has no dependence on a smallest positive weighted eigenvalue. We prove it on an arbitrary fixed graph, allowing several component matrices at a vertex.

Theorem 3.1. *Fix a finite graph with N vertices, rank $r \geq 2$, positive masses A_i, unitary edge matrices U_e, and a finite list of Hermitian matrices T_a at vertices $i(a)$. For arbitrary constant nonnegative weights w_e, β_a put*

$$\Phi_{w,\beta}(H) = \frac{1}{2}\sum_e w_e d(H_i, U_e H_j U_e^*)^2 + \sum_a \beta_a f_{T_a}(H_{i(a)}), \tag{13}$$

$$\mathcal{F}_{w,\beta}(H) = \sum_e w_e d(H_i, U_e H_j U_e^*) + \sum_a \beta_a \sqrt{2 f_{T_a}(H_{i(a)})}. \tag{14}$$

There are $R_* > 0$ and $c, C > 0$, depending only on the fixed data, such that, on $d_A(H, o) \leq R_*$,

$$c\mathcal{F}_{w,\beta}(H) \leq \|\operatorname{grad}\Phi_{w,\beta}(H)\|_{g_A} \leq C\mathcal{F}_{w,\beta}(H). \tag{15}$$

The negative gradient trajectory starting at distance $R \leq R_*$ exists for all $s \geq 0$, stays in that ball, and satisfies

$$\int_0^\infty \big(\mathcal{F}_{w,\beta}(H(s)) + \|\dot{H}(s)\|_{g_A}\big)\, ds \leq CR, \qquad \int_0^\infty \Phi_{w,\beta}(H(s))\, ds \leq \frac{1}{2}R^2. \tag{16}$$

It converges to a minimizer. These constants are independent of all weights, their ratios and the subset of weights that vanish.

3.1. **A weighted linear estimate.** The elementary ingredient is a formula for a weighted least-squares left inverse. Its constant retains the geometry of the fixed rows.

Lemma 3.2. *For every fixed real matrix D there is $C_D < \infty$ such that*

$$\|WDx\| \leq C_D \|D^T W D x\| \tag{17}$$

for every nonnegative diagonal matrix W and every x.

Proof. First suppose D is an $m \times n$ matrix of full column rank and W is positive. Let $D_S = R_S D$ be its restriction to n rows, and let $\mathfrak{B}$ consist of the subsets for which $\det D_S \neq 0$. Set $G = D^T W D$. Cauchy–Binet gives

$$\det G = \sum_{|S|=n} \det(D_S)^2 \prod_{j \in S} w_j.$$

In particular the numbers

$$p_S = \frac{\det(D_S)^2 \prod_{j \in S} w_j}{\det G} \quad (S \in \mathfrak{B})$$

are nonnegative and sum to one. Replacing column k of D by y replaces column k of $D^T W D$ by $D^T W y$. Cauchy–Binet for this replacement, followed by Cramer's rule, gives

$$G^{-1} D^T W = \sum_{S \in \mathfrak{B}} p_S D_S^{-1} R_S. \tag{18}$$

Terms with singular D_S already vanish before Cramer's rule is applied. Each R_S has norm one, so the norm of (18) is at most $\max_{S \in \mathfrak{B}} \|D_S^{-1}\|$. Transposing and applying the result to Gx proves (17).

For a matrix of rank $p > 0$, choose an isometry $Q : \mathbb{R}^p \to \mathbb{R}^n$ onto $(\ker D)^\perp$. Apply the preceding argument to $D' = DQ$. The identities $D = D'Q^T$ and $D^T = Q(D')^T$ preserve the norm on the right of (17). Finally replace W by $W + \delta I$ and let $\delta \downarrow 0$. This limit uses the inequality, not an inverse of a singular normal matrix. The case $D = 0$ is immediate. □

If D is divided into J fixed blocks D_k and W has constant weight $\lambda_k \geq 0$ on each block, the lemma gives

$$F^0(x) := \sum_k \lambda_k \|D_k x\| \leq \sqrt{J}\, C_D \|D^T W D x\|. \tag{19}$$

The choice of $(\ker D)^\perp$ is made once for the full fixed matrix. It does not change with the support of W.

3.2. **Exact zero sets and nonlinear absorption.** Use real coordinates $x_i = \sqrt{A_i} \log H_i$. Then $|x| = d_A(H, o)$ and the pullback metric is Euclidean at zero. The linear edge maps and component maps are

$$D_e x = \frac{x_i}{\sqrt{A_i}} - \mathrm{Ad}(U_e) \frac{x_j}{\sqrt{A_j}}, \qquad C_a x = i[T_a, x_{i(a)}/\sqrt{A_{i(a)}}].$$

Both maps take values in real spaces of Hermitian matrices. Associate to an edge the function $f_e = \frac{1}{2} d(H_i, U_e H_j U_e^*)^2$, and to a component matrix the function $f_a = f_{T_a}(H_{i(a)})$. The exact zero set of f_e is $\ker D_e$, and its Hessian at zero is $D_e^T D_e$.

For the component assertion, diagonalize $S = \log H$ with eigenvalues s_b. In this basis T is Hermitian and direct multiplication gives

$$f_T(e^S) = 2 \sum_{b<c} |T_{bc}|^2 \big(\cosh(s_b - s_c) - 1\big) \geq \frac{1}{2} \|[T, S]\|^2. \tag{20}$$

Its exact zero set is $[T, S] = 0$, and its quadratic term is $\frac{1}{2}\|[T, S]\|^2$. Thus the component Hessian is $C_a^T C_a$. Stack all these maps into D. Denote any one block by D_k and its associated function by f_k. Every f_k is nonnegative, vanishes precisely on $\ker D_k$, and has Hessian $D_k^T D_k$ at zero. In particular its gradient vanishes on the whole kernel.

On the auxiliary ball $|x| \le 1$ the third derivatives of this finite collection of functions are bounded. Put $r_k(x) = \nabla f_k(x) - D_k^T D_k x$. On $|x| \le R \le 1$, $\|Dr_k(x)\| \le C_0 R$. If P_k is orthogonal projection onto $\ker D_k$, then $r_k(P_k x) = 0$ and the segment from $P_k x$ to x stays in the ball. Therefore

$$\|r_k(x)\| \le C_0 R \operatorname{dist}(x, \ker D_k) \le C_1 R \|D_k x\|. \tag{21}$$

The last constant uses only the positive singular values of each nonzero fixed block. Zero blocks can be omitted. Combining (21) with (19), and writing $K = \sqrt{J} C_D$, gives

$$F^0(x) \le K \|\nabla \Phi_{w,\beta}(x)\| + K C_1 R F^0(x).$$

Choose R_* so that $K C_1 R_* \le \frac{1}{2}$. Absorption gives $F^0 \le 2K \|\nabla \Phi_{w,\beta}\|$ uniformly in the weights.

For edges, the distance-decreasing logarithm at the identity and the bounded differential of the exponential on this ball imply $\|D_e x\| \le \sqrt{2 f_e(x)} \le C \|D_e x\|$. For components, the lower bound is (20); Taylor expansion along the segment from $P_a x$ to x, on which the initial function and gradient vanish, gives the upper bound. Hence $F^0 \le \mathcal{F}_{w,\beta} \le C F^0$. Smooth equivalence of the coordinate metric with the Euclidean metric proves the left side of (15). The block bounds $\|\nabla f_k\| \le C \|D_k x\|$ prove the right side.

3.3. **Convexity and finite length.** To verify geodesic convexity of f_T, write $H(t) = H_0^{1/2} e^{tK} H_0^{1/2}$ and diagonalize the Hermitian matrix K with eigenvalues k_b. For $B = H_0^{-1/2} T H_0^{1/2}$,

$$\operatorname{tr}(T H(t)^{-1} T H(t)) = \sum_{b,c} |B_{cb}|^2 e^{t(k_b - k_c)}.$$

Each summand is convex. Squared distance between two moving geodesics is convex on a Hadamard manifold, and unitary conjugation is an isometry. Thus (13) is geodesically convex, nonnegative and minimized at o. Its negative gradient flow obeys

$$\frac{1}{2} \frac{d}{ds} d_A(H(s), o)^2 \le -\Phi_{w,\beta}(H(s)). \tag{22}$$

It stays in the compact ball of radius R and exists for all time. Integration gives the second estimate of (16).

For $T > s$, convexity and decreasing potential similarly give

$$\frac{1}{2} \frac{d}{ds} d_A(H(s), H(T))^2 \le \Phi_{w,\beta}(H(T)) - \Phi_{w,\beta}(H(s)) \le 0.$$

The curve is therefore self-contracted: its distance to any later point decreases as its earlier time increases. We apply Ohta's bounded-image length estimate on a finite-dimensional Hadamard manifold [Oht20, Theorem 3.3 and (3.8)]. Here the dimension is $N(r^2 - 1) \ge 3$, the image has diameter at most $2R$, and its radius-one neighborhood lies in $B(o, R_* + 1)$, whose volume is fixed. The length estimate consequently gives $\int_0^\infty \|\dot{H}\|_{g_A} \, ds \le CR$. Together with (15), this proves the first estimate of (16).

Finite length implies convergence to a point H_∞ in the ball. If its gradient were nonzero, continuity would give a positive lower bound for the speed for all sufficiently large times, a contradiction. Convexity makes H_∞ a global minimizer, so all terms with positive weights vanish there. In logarithm coordinates the minimizer set is exactly $\bigcap_{\lambda_k > 0} \ker D_k$. The limit need not be o, and the theorem gives neither a uniform rate nor an explicit formula for this selected limit. This completes the proof of Theorem 3.1.

3.4. **Dependence on the fixed constraints.** The fixed-matrix hypothesis cannot be replaced by uniform injectivity of the full matrix. For $\delta > 0$ take

$$D_\delta = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 1 & \delta \end{pmatrix}, \quad W_\delta = \operatorname{diag}(1, \delta^4, 1), \quad x_\delta = \begin{pmatrix} -\delta/2 \\ 1 \end{pmatrix}.$$

Although $D_\delta^T D_\delta \geq I$,

$$W_\delta D_\delta x_\delta = (-\delta/2, \delta^4, \delta/2)^T, \qquad D_\delta^T W_\delta D_\delta x_\delta = (0, \delta^2/2 + \delta^4)^T.$$

Their norm ratio is asymptotic to $\sqrt{2}/\delta$. A row basis that can dominate the weighted normal matrix becomes singular. This is an obstruction to a uniform force estimate, not an obstruction theorem for actual HYM approximation.

There is a second distinction needed in the analytic proof. A component force is weighted by α^2, whereas a change of component frame can have a first-order factor α. With one vertex, mass one, $T = \sigma_z$, $\kappa = 1$ and $H = \exp(q\sigma_x)$, direct calculation gives

$$f_T(H) = 4\sinh^2 q, \qquad g(\partial_q H, \partial_q H) = 2, \qquad \dot{q} = -2\alpha^2 \sinh(2q).$$

For $q(0) = q_0 > 0$ and $\alpha > 0$ it follows that

$$\int_0^\infty \alpha q(s)\, ds = \frac{1}{2\alpha} \int_0^{q_0} \frac{q}{\sinh(2q)}\, dq. \tag{23}$$

The right-hand integral is finite and strictly positive. Thus a first-order frame error cannot be integrated uniformly by Theorem 3.1 alone. The angular divergence construction below addresses precisely this term.

4. HYM comparison with a divergence residual

This section applies to any closed Riemann surface C with a fixed Kähler metric and any flat unitary bundle (E, D, h_0) with parallel determinant metric. A Hermitian metric is represented by a positive h_0-self-adjoint endomorphism. The connection D, its formal adjoint D^*, and the norms of endomorphism-valued forms always use this fixed reference metric. Local parallel unitary frames have constant unitary transition maps.

4.1. **The Chern equation and its trace inequality.** With area form $dA = a(z)\, dx\, dy = (i/2)a(z)\, dz \wedge d\bar{z}$, put $\Delta = -4a^{-1}\partial_z\partial_{\bar{z}}$, the nonnegative scalar Laplacian. The equation

$$h^{-1}h_s = -2i\Lambda F_h \tag{24}$$

in a local parallel holomorphic frame is

$$h_s = 4a^{-1}(h_{z\bar{z}} - h_{\bar{z}}h^{-1}h_z), \qquad 2i\Lambda F_h = -4a^{-1}\partial_{\bar{z}}(h^{-1}h_z). \tag{25}$$

In real coordinates it reads

$$\begin{aligned} ah_s = {} & h_{xx} + h_{yy} - h_x h^{-1} h_x - h_y h^{-1} h_y \\ & + i(h_x h^{-1} h_y - h_y h^{-1} h_x). \end{aligned} \tag{26}$$

The last, Hermitian commutator term is retained throughout the proof. For $h(x, y) = e^{xV+yW}$ at the origin, with noncommuting Hermitian V, W, the first four terms cancel and this term is $i[V, W]$. Thus (24) is not ordinary harmonic-map heat flow into $\mathcal{P}_r$.

For smooth positive initial data, the classical compact-base HYM existence theorem gives a smooth solution for every finite time [Don85]; a precise arbitrary-initial-metric statement and continuation proof are in [Kob87, Theorem 6.8.25]. The book uses time $t = 2s$ in our

normalization; the statement in [HKKP18, §4.1, following (4.8)] uses the normalization above. The scalar equation for $\log\det h$ preserves the parallel determinant. The maximum principle applied to v^*hv for a constant local vector v gives the upper order bound: the quadratic term in (25) is nonpositive after this pairing. For h^{-1} the inverse derivative formula gives

$$(h^{-1})_s = 4a^{-1}\big((h^{-1})_{z\bar z} - (h^{-1})_z h (h^{-1})_{\bar z}\big),$$

which gives its upper bound as well. Hence

$$ch_0 \le h(0) \le C_0 h_0 \quad\Longrightarrow\quad ch_0 \le h(s) \le C_0 h_0 \quad (s \ge 0). \tag{27}$$

These order comparisons are global and independent of spatial derivatives of the initial metric. Existence is asserted separately on each fixed compact smooth fibre; no uniform high-derivative continuation theorem on degenerating fibres is needed.

Define the trace distance and the residual of a comparison metric by

$$D_{\mathrm{tr}}(h,K) = \operatorname{tr}(h^{-1}K + K^{-1}h - 2I), \qquad \mathcal{R}(K) = K^{-1}K_s + 2i\Lambda F_K. \tag{28}$$

If the eigenvalues of $h^{-1}K$ are e^{b_j}, then

$$d(h,K)^2 = \sum_j b_j^2 \le \sum_j (e^{b_j} + e^{-b_j} - 2) = D_{\mathrm{tr}}(h,K). \tag{29}$$

On a fixed bounded positive range the reverse inequality holds with a fixed constant.

Lemma 4.1. *For an exact solution h and any smooth comparison metric K,*

$$(\partial_s + \Delta) D_{\mathrm{tr}}(h,K) \le \operatorname{Re}\operatorname{tr}\big((h^{-1}K - K^{-1}h)\mathcal{R}(K)\big). \tag{30}$$

Proof. First take two exact flows. At a specified point and time choose a time-independent holomorphic frame normal for h, so $h = I$ and $\partial h = 0$ at that point. Substitution of the Chern equation into the time derivative and scalar Laplacian of $\operatorname{tr}(h^{-1}K)$ cancels the curvature terms. The remaining expression is $-4a^{-1}\operatorname{tr}(K_{\bar z}K^{-1}K_z) \le 0$. Interchanging the two metrics proves the inequality for the inverse trace. For a forced comparison, its extra velocity is $K\mathcal{R}(K)$. Its contributions to these two traces are respectively $\operatorname{tr}(h^{-1}K\mathcal{R}(K))$ and $-\operatorname{tr}(\mathcal{R}(K)K^{-1}h)$. This proves (30). The real part is redundant for the actual residual because $K\mathcal{R}(K)$ is Hermitian; retaining it allows complex pieces in a decomposition of that residual. The calculation is invariant under changes of holomorphic frame and so defines a global scalar inequality. □

4.2. **A comparison estimate using energy.** For an interval $[0,S]$ write

$$E_h(S) = \int_0^S \int_C |Dh|^2\, dA\, ds, \qquad E_K(S) = \int_0^S \int_C |DK|^2\, dA\, ds.$$

The following lemma uses an actual global decomposition of the residual. It does not assert that any small local error admits one.

Lemma 4.2. *Suppose h, K have determinant equal to that of h_0 and satisfy $cI \le h, K \le C_0 I$. Suppose, in distributions on the closed surface,*

$$\mathcal{R}(K) = D^*J + B, \tag{31}$$

where J is a complex endomorphism-valued one-form and B an endomorphism-valued function. Put

$$J_2(S)^2 = \int_0^S \int_C |J|^2, \qquad B_1(S) = \int_0^S \int_C |B|, \qquad \eta = \int_C D_{\mathrm{tr}}(h(0),K(0)).$$

For every $S > 0$ for which these quantities are finite,

$$\sup_{0\le s\le S}\int_C d(h(s),K(s))^2 \le \eta + CB_1(S) + C\big(E_h(S)^{1/2} + E_K(S)^{1/2}\big)J_2(S). \tag{32}$$

Moreover

$$E_h(\infty) \le C\int_C \operatorname{tr}(h(0)-I)\,dA \le C'\operatorname{Area}(C). \tag{33}$$

The constants depend only on c, C_0, r. It is enough in (31) that J be piecewise H^1 with zero normal traces at its internal boundaries, extended by zero elsewhere. No spectral gap enters these estimates.

Proof. Put $Z = h^{-1}K - K^{-1}h$. On the bounded matrix range $|Z| \le C$. The reference connection obeys the inverse and product rules, so

$$DZ = -h^{-1}(Dh)h^{-1}K + h^{-1}DK + K^{-1}(DK)K^{-1}h - K^{-1}Dh. \tag{34}$$

In particular $|DZ| \le C(|Dh| + |DK|)$. Compatibility with trace multiplication gives the bilinear integration-by-parts formula

$$\int_C \operatorname{Re}\operatorname{tr}(ZD^*J) = \int_C \operatorname{Re}\operatorname{tr}((DZ)\cdot J). \tag{35}$$

The dot contracts real cotangent indices; it does not reverse the order of the matrices or insert an adjoint. Equivalently one can apply the Hermitian formal-adjoint identity with test field Z^*. For the stated piecewise fields, integration over each piece gives the same formula because the normal traces vanish. Tangential jumps do not contribute to divergence. More generally this is the defining weak pairing whenever (31) holds with the declared integrability.

Integrate (30) on the closed surface and in time. The scalar Laplacian integrates to zero. Use (35), (34) and Cauchy–Schwarz in space-time for the two derivative terms. The B term is at most $CB_1(S)$. This first proves the bound for the trace distance; (29) yields (32).

For the energy estimate, the exact equation (25) gives

$$-\frac{d}{ds}\int_C \operatorname{tr}(h-I)\,dA = 4\int_C \operatorname{tr}(h_z^* h^{-1} h_z)\,dx\,dy. \tag{36}$$

The derivative term integrates to zero; the displayed ordered quadratic term is nonnegative. Only now replace h^{-1} by its scalar lower bound. The result controls a fixed multiple of $\int_C |Dh|^2\,dA$. Determinant one implies $\operatorname{tr}(h-I) \ge 0$ by the arithmetic–geometric mean inequality. Integrating to S and then letting $S \to \infty$ proves (33). It requires neither convergence to a particular parallel metric nor a bound on the initial spatial derivative. □

In particular, on a bounded-area family, all-time approximation follows if the initial trace error tends to zero, $E_K(\infty)$ stays bounded, and $J_2(\infty), B_1(\infty)$ tend to zero. The remainder of the proof constructs exactly these quantities on the compact family of Section 2.

5. The global comparison metric

Return to the family of Section 2. All parameters are constant during a trajectory. Let $H(s)$ solve (12) in a fixed product ball about o. Put

$$\alpha_* = \max(|\alpha_+|, |\alpha_-|), \qquad \rho = \epsilon_*, \qquad \ell = 1 + |\log\rho|, \qquad \sigma = \alpha_*^2 + \epsilon_*. \tag{37}$$

We always take $\alpha_* \to 0$ and $L_* \to \infty$, with no restriction on their ratios or the individual L_e. Constants in this section depend on the fixed component data and product ball. They can

be chosen uniformly over unitary transports. The physical speed is $O(\sigma)$; its product with L_* need not be bounded.

For later estimates define

$$d_e = d(H_i, U_{ij} H_j U_{ij}^*), \qquad F_{\mathrm{ed}} = \sum_e \epsilon_e d_e, \qquad E_{\mathrm{ed}} = \sum_e \epsilon_e d_e^2, \tag{38}$$

$$m_i = \|[T_i, H_i]\|, \qquad F_{\mathrm{co}} = \sum_{i=+,-} \alpha_i^2 \kappa_i \sqrt{2 f_{T_i}(H_i)}, \qquad V = \|\dot{H}\|_{g_A}. \tag{39}$$

Write $F = F_{\mathrm{ed}} + F_{\mathrm{co}}$. Fixed factors 2π are absorbed in constants. Smoothness on the bounded matrix range and (20) give

$$\begin{gathered} V \le CF, \qquad F_{\mathrm{ed}} \le C\epsilon_*, \qquad |\dot{B}_{ij}| \le CV, \\ F_{\mathrm{ed}}^2 \le C\epsilon_* E_{\mathrm{ed}}, \qquad \sum_i \alpha_i^2 m_i^2 + E_{\mathrm{ed}} \le C\Phi(H), \qquad \sum_i \alpha_i^2 m_i \le C F_{\mathrm{co}}. \end{gathered} \tag{40}$$

For example $[T, e^S] = \int_0^1 e^{(1-t)S}[T,S]e^{tS}\,dt$ bounds m_i by $C\|[T_i, \log H_i]\|$. The finite number of edges gives the first inequality on the second line by Cauchy–Schwarz.

5.1. **Component Green functions and node profiles.** Let g_{ie} be the mean-zero Green function on the fixed component C_i with

$$\Delta_i g_{ie} = -2\pi\delta_{p_{ie}} + 2\pi/A_i, \qquad g_{ie}(u) = \log|u| + \gamma_{ie}(u). \tag{41}$$

In the harmonic unitary gauge put

$$\xi_i = -\sum_{e=ij} \epsilon_e B_{ij} g_{ie}, \qquad K_i = H_i e^{\xi_i}, \qquad \Delta_i \xi_i = -\frac{2\pi}{A_i} \sum_{e=ij} \epsilon_e B_{ij} \tag{42}$$

off the punctures. The matrices ξ_i are H_i-self-adjoint and trace-free. On the component cut off at radius $\rho/2$, the fixed Green functions give

$$\begin{gathered} \|\xi_i\|_\infty \le C\ell F_{\mathrm{ed}}, \qquad \int (|\xi_i| + |\nabla\xi_i|) \le C F_{\mathrm{ed}}, \qquad \int |\xi_i|^2 \le C F_{\mathrm{ed}}^2, \\ \int |\nabla\xi_i|^2 \le C\ell F_{\mathrm{ed}}^2, \qquad \int |\xi_i||\Delta_i\xi_i| \le C F_{\mathrm{ed}}^2, \qquad \int |\partial_s \xi_i| \le C\epsilon_* V. \end{gathered} \tag{43}$$

The logarithmic loss comes only from the integral of $|\nabla \log|u||^2$.

At incidence ie expand

$$\xi_i(u) = -\epsilon_e B_{ij} \log|u| + c_{ie} + q_{ie}(u), \qquad q_{ie}(0) = 0, \qquad P_{ie} = H_i e^{c_{ie}}. \tag{44}$$

The constant c_{ie} is the regular part of the complete Green sum, including all other punctures of the component. It satisfies $|c_{ie}| \le C F_{\mathrm{ed}}$, and on $\rho \le |u| \le 2\rho$,

$$|\nabla^k q_{ie}| \le C F_{\mathrm{ed}} \rho^{1-k} \quad (0 \le k \le 2). \tag{45}$$

The corresponding time derivatives have F_{ed} replaced by $\epsilon_* V$.

In the node frame at its i end define

$$N_e(s,u) = \mathrm{Geo}(P_{ie}, U_{ij} P_{je} U_{ij}^*; \theta_e), \qquad \theta_e = -\epsilon_e \log|u|. \tag{46}$$

Here $\mathrm{Geo}(P,Q;\theta) = P\exp(\theta \log(P^{-1}Q))$. The expression from the opposite end gives the same metric: the parameter becomes $1-\theta_e$ because $uv = \tau_e$ and $\epsilon_e L_e = 1$. This identity requires the individual ϵ_e, rather than the largest ϵ_*. For each fixed time, $N_e^{-1}\partial N_e$ is a constant matrix times $\partial \log|u|$, so its Chern curvature vanishes on the annulus. Its endpoint velocities are at most CV.

Let $\chi(|u|/\rho)$ be a smooth radial cutoff, zero near the inner circle of the collar $\rho < |u| < 2\rho$ and one near the outer circle. The component metric in the same flat node frame is $G_{ie}^* K_i G_{ie}$. On the collar set

$$\widehat{K} = H_i \exp\big((1-\chi)\log(H_i^{-1}N_e) + \chi \log(H_i^{-1}G_{ie}^* K_i G_{ie})\big). \tag{47}$$

Use K_i on the component exterior and N_e on the middle node. All logarithms in (47) are H_i-self-adjoint and trace-free. These formulas therefore give a smooth positive metric on the actual bundle, with the same determinant as h_0. They are equivariant under the constant unitary node transitions. Since $\epsilon_*\ell \to 0$, the component corrections are small and the comparison metrics have fixed two-sided bounds for all times. On every fixed compact component region,

$$d(\widehat{K}(H), H_i) \le C\epsilon_*. \tag{48}$$

5.2. **The full component curvature calculation.** For spatially constant H_i in the harmonic gauge, put $B_i = H_i^{-1}T_iH_i$. The Chern connection satisfies

$$H_i^{-1}\partial_{D_i}H_i = -i\alpha_i(B_i - T_i)\vartheta_i^{1,0}, \qquad F_{H_i} = \alpha_i^2[T_i, B_i]\vartheta_i^{1,0}\wedge\vartheta_i^{0,1}.$$

With our metric convention, $2i\Lambda(\vartheta_i^{1,0}\wedge\vartheta_i^{0,1}) = |\vartheta_i|^2 = \kappa_i/A_i$. Thus

$$\mathcal{C}_i(H_i) := 2i\Lambda_i F_{H_i} = \frac{\alpha_i^2\kappa_i}{A_i}[T_i, H_i^{-1}T_iH_i]. \tag{49}$$

Equations (12) and (42) give the exact leading cancellation

$$H_i^{-1}\dot{H}_i + \mathcal{C}_i(H_i) + \Delta_i\xi_i = 0. \tag{50}$$

We record how the complete Chern expression is controlled, since its ordered terms are needed on an unbounded time interval. The identity

$$K_i^{-1}\partial_{D_i}K_i = e^{-\xi_i}(H_i^{-1}\partial_{D_i}H_i)e^{\xi_i} + e^{-\xi_i}\partial_{D_i}e^{\xi_i}$$

and the exact formula

$$e^{-\xi}\partial e^{\xi} = \int_0^1 e^{-t\xi}(\partial\xi)e^{t\xi}\,dt$$

imply, after covariant differentiation,

$$|2i\Lambda_i F_{K_i} - \mathcal{C}_i(H_i) - \Delta_i\xi_i| \le C\big(\alpha_*|\nabla\xi_i| + \alpha_*^2|\xi_i| + |\nabla\xi_i|^2 + |\xi_i||\Delta_i\xi_i|\big). \tag{51}$$

Indeed the first two errors have respectively one connection coefficient and a derivative of ξ_i, or two connection coefficients and ξ_i. The coefficients of ϑ_i are constant in flat coordinates. The remaining errors come from the exponential integral and contain two derivatives of ξ_i or its product with its Laplacian. Further factors of ξ_i are absorbed by its small supremum norm. On rational components $\alpha_i = 0$. The time identity bounds $|K_i^{-1}\partial_s K_i - H_i^{-1}\dot{H}_i|$ by $C(|\partial_s\xi_i| + V|\xi_i|)$.

Combining these statements with (43) gives the component residual bound

$$\begin{aligned}\int |\mathcal{R}(K_i)| &\le C\big(\epsilon_* V + VF_{\mathrm{ed}} + \ell F_{\mathrm{ed}}^2 + \alpha_* F_{\mathrm{ed}} + \alpha_*^2 F_{\mathrm{ed}}\big)\\ &\le C(\alpha_* + \epsilon_*\ell)(F+V).\end{aligned} \tag{52}$$

The branch identification preserves both the complex structure and the reference connection exactly. Transfer to the fibre changes only the metric contraction and area. Lemma 2.2 adds at most

$$Cb^2\rho^{-N}(F+V) \tag{53}$$

for one fixed finite N. To retain this force factor, note that the constant-metric curvature in (49) is bounded by $C\alpha_i^2 m_i$; every other spatial term contains ξ_i or its derivatives. The latter lose only fixed powers of ρ^{-1}. The middle-node time residual has integral at most $C\rho^2 V$, since its spatial curvature vanishes and its area is $O(\rho^2)$. These estimates already have the desired integrable form on a small finite basin. The incidence collars require the additional argument in the next section.

5.3. **Energy and dependence on the finite tuple.** The comparison metric has an integrated energy bound even without the small-ball force estimate.

Lemma 5.1. *On every fixed finite ball, for sufficiently small joint parameters,*

$$\int_{C_\tau} |D\widehat{K}(H)|^2 \le C\Phi(H), \qquad \int_0^\infty \int_{C_\tau} |D\widehat{K}(H(s))|^2 \le C d_A(H(0),o)^2. \tag{54}$$

Also the map $H \mapsto \widehat{K}(H)$ is uniformly Lipschitz from that ball to the integrated distance on C_τ:

$$\int_{C_\tau} d(\widehat{K}(H),\widehat{K}(H'))^2 \le C d_A(H,H')^2. \tag{55}$$

Constants are uniform in the unitary transports and all scale ratios.

Proof. The component derivative is covariant with respect to $D_i = d - i\alpha_i[T_i,\cdot]\vartheta_i$. From (43) and bounded differentials of the exponential,

$$\int |D_i K_i|^2 \le C\big(\alpha_i^2 m_i^2 + (\ell+\alpha_i^2)F_{\mathrm{ed}}^2\big). \tag{56}$$

Real one-form energy on a Riemann surface is conformally invariant, so the exact holomorphic identification transfers this inequality to the fibre without a metric error. Its sum is bounded by $C\Phi$ using (40) and $\epsilon_*\ell \to 0$.

The two corrected endpoints at edge e have distance at most $d_e + CF_{\mathrm{ed}}$. Constant geodesic speed in (46) gives middle-node energy at most

$$C\epsilon_e^2(L_e + 2\log\rho)(d_e+F_{\mathrm{ed}})^2 \le C\epsilon_e(d_e+F_{\mathrm{ed}})^2.$$

Its sum is again bounded by $C\Phi$.

On a collar, let δ_{ie} be the difference of the two relative logarithms in (47). Differentiating $G_{ie}^* H_i G_{ie}$ first retains the factor m_i in its first derivative. The action on ξ_i contributes the other gauge term below. The regular Green part and the mixed endpoint/geodesic expansion give

$$\begin{aligned} |\delta_{ie}| + \rho|\nabla\delta_{ie}| &\le C a_{ie}, \\ a_{ie} &= |\alpha_i|\rho m_i + |\alpha_i|\rho\ell F_{\mathrm{ed}} + \rho F_{\mathrm{ed}} + \epsilon_e \ell F_{\mathrm{ed}}. \end{aligned} \tag{57}$$

For the last term, at zero endpoint corrections the node logarithm is exactly $\theta_e B_{ij}$; at $\theta_e = 0$ it is c_{ie}. The remainder from their sum vanishes on both axes and is at most $C\theta_e F_{\mathrm{ed}}$, with the corresponding derivative bounds. Since $\theta_e \le C\epsilon_e\ell$ on the collar, this gives the displayed term. The derivative of the radial node logarithm is at most $C\rho^{-1}\epsilon_e(d_e+F_{\mathrm{ed}})$. Consequently the collar energy of the interpolated logarithms is at most

$$C\big(\epsilon_e^2(d_e+F_{\mathrm{ed}})^2 + a_{ie}^2\big).$$

Here the cutoff derivative contributes ρ^{-1} and the Euclidean collar area contributes $O(\rho^2)$. Every term is bounded by $C\Phi$ after summing incidences: use (40), $\rho = \epsilon_*$ and bounded $\epsilon_*\ell$.

This proves the instantaneous energy bound. Integrating (22) proves the second bound of (54).

For (55), the endpoint/logarithm maps have bounded derivatives on the finite ball. Component Green functions have bounded L^2 norms and coefficients ϵ_e. Their supremum on the truncated component is bounded after multiplication by ϵ_e. On middle nodes $0 \le \theta_e \le 1$, so geodesic endpoint dependence is bounded; their total area is $O(\rho^2)$. The same smooth endpoint dependence and the bounded collar logarithms apply to (47). Integrating along a finite geodesic from H to H' and using Cauchy–Schwarz proves the stated estimate. □

6. Angular cancellation and all-time approximation

We now fix the transports U_e and component matrices T_i, and restrict the initial finite tuple to the ball supplied by Theorem 3.1. In addition to the energy bounds of Section 5, we then have

$$\int_0^\infty (F+V)\,ds \le C, \qquad \int_0^\infty \Phi(H(s))\,ds \le C. \tag{58}$$

The constants may depend on the fixed constraint maps. We use the same reconstructed metric $\widehat{K}$; the argument changes how its residual is estimated.

Proposition 6.1. *There are global complex endomorphism-valued one-forms $J(s)$ and integrable endomorphisms $B(s)$ such that*

$$\mathcal{R}(\widehat{K}) = D^*J + B, \qquad \int_0^\infty \int_{C_\tau} |J|^2 \le C\rho^2, \qquad \int_0^\infty \int_{C_\tau} |B| \le C a_*, \tag{59}$$

where, for one fixed finite exponent N,

$$a_* = \alpha_* + \epsilon_* \ell^2 + b^2 \rho^{-N} \longrightarrow 0. \tag{60}$$

The divergence identity is distributional on the closed fibre. The forms J are smooth on each incidence collar, have zero normal trace on both boundary circles, and are extended by zero elsewhere. Constants are independent of time and every ratio among the component and node scales.

6.1. **A radial reference metric.** Consider an elliptic incidence collar $A_\rho = \{\rho < |z| < 2\rho\}$ in its parallel node frame. At the current time keep the finite tuple, all weights and the corrected node endpoints fixed. Introduce an auxiliary real variable a and set

$$G_a(z) = e^{iaT_iF_{ie}(z)}, \qquad \xi_i^{\mathrm{rad}} = -\epsilon_e B_{ij} \log|z| + c_{ie}.$$

The variable a is used only for a spatial Taylor expansion. It does not change the trajectory or any parameter along that trajectory. Define

$$K_a^{\mathrm{rad}} = H_i \exp\Big((1-\chi)\log(H_i^{-1}N_e) + \chi \log\{H_i^{-1} G_a^* H_i e^{\xi_i^{\mathrm{rad}}} G_a\} \Big). \tag{61}$$

The metric K_0^{rad} is radial. The actual collar metric is obtained at $a = \alpha_i$ by replacing ξ_i^{rad} with $\xi_i^{\mathrm{rad}} + q_{ie}$. No matrix factors have been commuted in this formula.

Write $\mathcal{C}(K) = 2i\Lambda_\tau F_K$ for the spatial Chern expression. Bounded derivatives of relative logarithms and exponentials, together with (45), imply

$$\int_{A_\rho} |\mathcal{C}(\widehat{K}) - \mathcal{C}(K_{\alpha_i}^{\mathrm{rad}})|\,dA_\tau \le C\rho F_{\mathrm{ed}}. \tag{62}$$

Indeed the change in each metric derivative of order $k \le 2$ is at most $CF_{\mathrm{ed}}\rho^{1-k}$. All other spatial inputs have derivatives bounded by $C\rho^{-k}$. Substitution into the full expression $-4a_\tau^{-1}\partial_{\bar z}(K^{-1}K_z)$ and multiplication by dA_τ give the asserted integral. The quadratic error is smaller on the bounded range.

At $a = 0$, comparison with the exactly flat node profile gives

$$\int_{A_\rho} |\mathcal{C}(K_0^{\mathrm{rad}})|\, dA_\tau \le C\epsilon_*\ell^2 F_{\mathrm{ed}}. \tag{63}$$

Here is the endpoint expansion used in this estimate. At zero regular endpoint corrections the relative node logarithm is $\theta_e B_{ij}$, whereas at $\theta_e = 0$ it is c_{ie}. The difference from their sum therefore has size at most $C\theta_e F_{\mathrm{ed}}$, with two spatial derivatives bounded by the corresponding factors ρ^{-1} and ρ^{-2}. On the collar $\theta_e = O(\epsilon_e\ell)$. In the curvature difference, the terms are bounded by

$$C\big(|\nabla^2\delta| + |\nabla n||\nabla\delta| + |\delta|(|\nabla^2 n| + |\nabla n|^2) + |\nabla\delta|^2\big),$$

where n is the node logarithm. Its derivatives of positive order are at most $C\epsilon_e\rho^{-k}$ on the bounded finite ball. Integrating over the collar gives (63). All time derivatives of the actual collar metric are bounded by CV; their residual integral is at most $C\rho^2 V$.

6.2. **The first variation has zero angular mean.** In a flat holomorphic frame with $dA_\tau = a_\tau(z)\, dx\, dy$,

$$\mathcal{C}(K) = -4a_\tau^{-1}\partial_{\bar z}(K^{-1}\partial_z K).$$

Put $K_0 = K_0^{\mathrm{rad}}$, $A_0 = K_0^{-1}\partial_z K_0$ and $W = K_0^{-1}\partial_a K_a^{\mathrm{rad}}|_{a=0}$. The inverse derivative rule gives the exact variation

$$\partial_a \mathcal{C}(K_a^{\mathrm{rad}})\Big|_0 = -4a_\tau^{-1}\partial_{\bar z}\{\partial_z W + [A_0, W]\}. \tag{64}$$

All inputs of (61) except G_a are radial. Its first variation thus has the form $W(z) = F_{ie}(z)W_0(|z|)$ for a radial matrix function W_0. By (7), W has only angular indices $1, -1$.

Multiplication of (64) by a_τ gives a rotationally invariant operator on the radial background. To see the matrix order explicitly, set $t = \log|z|$ and $C(t) = K_0^{-1}\partial_t K_0$. If $W = e^{in\theta}w(t)$, direct differentiation gives

$$\begin{aligned} 4\partial_{\bar z}\{\partial_z W + [A_0, W]\} = e^{-2t+in\theta}\big(&w'' - n^2 w + [C', w] \\ &+ [C, w'] - n[C, w]\big). \end{aligned} \tag{65}$$

In particular each angular index is preserved, even when the matrices fail to commute. Therefore the matrix-valued function

$$S_{ie} = a_\tau\alpha_i\, \partial_a \mathcal{C}(K_a^{\mathrm{rad}})\Big|_0 \tag{66}$$

has only the two indices $1, -1$, and

$$\int_{A_\rho} S_{ie}\, dx\, dy = 0 \tag{67}$$

entry by entry. The factor a_τ in (66) is essential. We have not assumed that the actual fibre metric is radial.

We next retain the force factors in this variation. For $0 \le k \le 2$, uniformly between $a = 0$ and $a = \alpha_i$,

$$\begin{aligned} |\nabla^k \partial_a K_a^{\mathrm{rad}}| &\le C\rho^{1-k}(m_i + \ell F_{\mathrm{ed}}), \\ |\nabla^k \partial_a^2 K_a^{\mathrm{rad}}| &\le C\rho^{2-k}(m_i + \ell F_{\mathrm{ed}}). \end{aligned} \tag{68}$$

To prove these estimates, first differentiate $G_a^* H_i e^{\xi_i^{\mathrm{rad}}} G_a$. Its first derivative is F_{ie} times a commutator with iT_i, and its second is F_{ie}^2 times an iterated commutator. The matrix factors are bounded by $C(m_i + |\xi_i^{\mathrm{rad}}|)$. Spatial derivatives differentiate the linear harmonic function, the radial cutoff or the radial logarithms. We have $|\xi_i^{\mathrm{rad}}| \le C\ell F_{\mathrm{ed}}$ and $|\nabla^k \xi_i^{\mathrm{rad}}| \le C F_{\mathrm{ed}} \rho^{-k}$ for $k = 1, 2$. The node logarithm has the same derivative bound because $\epsilon_e(d_e + F_{\mathrm{ed}}) \le C F_{\mathrm{ed}}$. Composition with relative logarithms and exponentials preserves these bounds. Products of two first parameter derivatives contain $(m_i + \ell F_{\mathrm{ed}})^2$, which is bounded by a fixed multiple of $m_i + \ell F_{\mathrm{ed}}$ on the finite ball. This proves (68), including when $m_i = 0$.

Using these estimates in the full Chern expression, with the inverse-matrix derivatives in their stated order, gives

$$\|S_{ie}\|_{L^2(dx\,dy)} \le C|\alpha_i|(m_i + \ell F_{\mathrm{ed}}). \tag{69}$$

Taylor's formula in a also gives

$$\int_{A_\rho} \left| \mathcal{C}(K_{\alpha_i}^{\mathrm{rad}}) - \mathcal{C}(K_0^{\mathrm{rad}}) - \alpha_i\, \partial_a \mathcal{C}(K_a^{\mathrm{rad}})\Big|_0 \right| dA_\tau \le C\alpha_i^2 \rho^2 (m_i + \ell F_{\mathrm{ed}}). \tag{70}$$

In each second parameter derivative of curvature there is either one second metric variation or two first variations. Two spatial derivatives in (68) bound both types. The factor a_τ^{-1} cancels against area. Thus this estimate includes the noncommuting quadratic terms, rather than estimating only the scalar principal part.

6.3. **Neumann primitives on the actual bundle.** For each real and imaginary matrix entry solve

$$-\Delta_{\mathbb{R}^2} u_{ie} = S_{ie}, \qquad \partial_n u_{ie} = 0 \text{ on both boundary circles}, \qquad \int_{A_\rho} u_{ie}\, dx\, dy = 0. \tag{71}$$

Equation (67) is exactly the compatibility condition. Minimizing the Dirichlet quadratic functional on mean-zero H^1 functions proves existence and uniqueness. Scaling the fixed annulus $\{1 < |z| < 2\}$ gives $\|u\|_2 \le C\rho \|du\|_2$. Testing (71) by u then gives

$$\|du_{ie}\|_{L^2(dx\,dy)} \le C\rho \|S_{ie}\|_{L^2(dx\,dy)}. \tag{72}$$

Set $J_{ie} = du_{ie}$ in the parallel node frame, and extend it by zero outside the collar. In real dimension two, $D^* = a_\tau^{-1} d^*_{\mathrm{Eucl}}$ in this frame. Testing against a smooth global endomorphism and integrating on the collar shows that its distributional divergence is precisely $\alpha_i \partial_a \mathcal{C}(K_a^{\mathrm{rad}})|_0$ there and zero elsewhere. There is no boundary measure, since the normal component vanishes on both circles. Tangential jumps of the extension do not enter divergence. Constant unitary changes of node frame conjugate this construction and give a global endomorphism-valued one-form.

Incidence collars are disjoint for sufficiently small parameters. Their one-form energy is conformally invariant. Hence, for $J = \sum_{ie} J_{ie}$,

$$\int_{C_\tau} |J|^2 \le C\rho^2 \sum_i \alpha_i^2 (m_i^2 + \ell^2 F_{\mathrm{ed}}^2) \le C\rho^2 \Phi(H). \tag{73}$$

The last inequality follows from (40) and boundedness of $\alpha_*^2 \epsilon_* \ell^2$ for small joint parameters. Rational incidences have $\alpha_i = 0$ and require no primitive. Each Neumann problem is on a fixed-shape collar. Its constant does not depend on any of the long middle annuli.

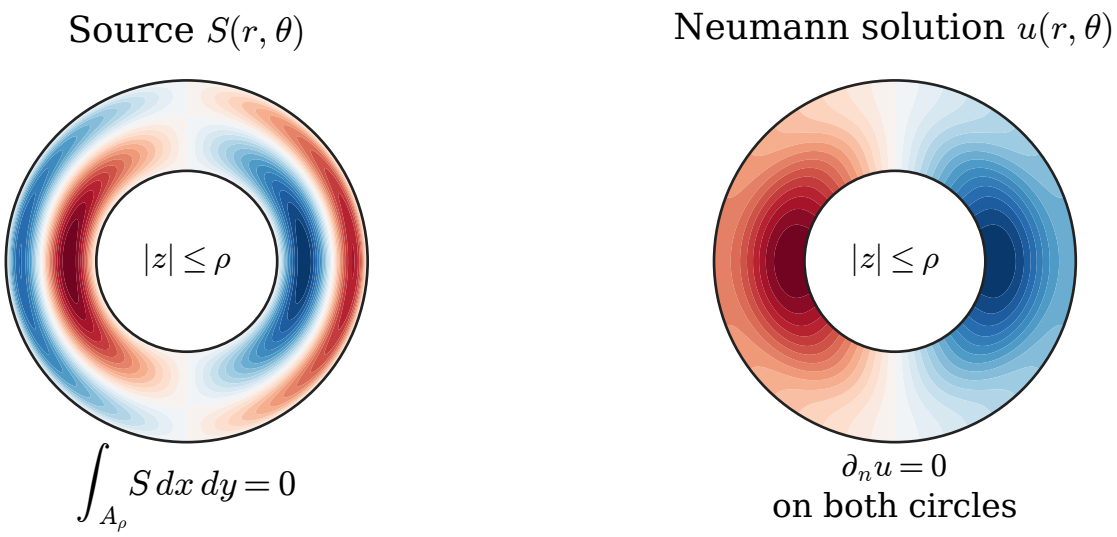


FIGURE 2. A scalar first-angular-mode example of the collar construction, with colors normalized separately in the two panels. For $x = r/\rho$, let $\chi(x) = 6(x-1)^5 - 15(x-1)^4 + 10(x-1)^3$ and $u = \rho(\chi(x)x - 4x/3 - 4/(3x))\cos\theta$. Then $S = -\Delta_{\mathbb{R}^2} u$ has zero mean and $\partial_n u = 0$ at $r = \rho, 2\rho$. The matrix construction uses the same compatibility condition and energy estimate; it does not require this particular scalar source.

Define $B = \mathcal{R}(\widehat{K}) - D^* J$ on the whole fibre. The component and middle-node estimates (52)–(53), the regular and radial collar estimates (62)–(63), and (70) now give

$$\int_{C_\tau} |B| \le C(\alpha_* + \epsilon_* \ell^2 + b^2 \rho^{-N})(F + V). \tag{74}$$

For the remaining Taylor terms use $\alpha_i^2 \rho^2 m_i \le C\rho^2 F_{\mathrm{co}}$ and $\alpha_i^2 \rho^2 \ell F_{\mathrm{ed}} \le C\alpha_*^2 \rho^2 \ell F_{\mathrm{ed}}$. The time terms are bounded by $C\rho^2 V$. Thus no term weighted only by $|\alpha_i| m_i$ remains in (74). Integrating (73) by the finite energy inequality and (74) by (58) proves Proposition 6.1.

6.4. **The all-time theorem.**

Theorem 6.2. *Fix the component geometry, rank, harmonic forms, component matrices and transports of Section 2. There is a radius $R_* > 0$ such that the following holds. Let the four nonzero node parameters and the two real component parameters tend independently to zero, and keep them fixed during each trajectory. Let $H(s)$ solve (12) from $d_A(H(0), o) \le R_*$. For any smooth determinant-normalized actual initial metric with fixed bounds $ch_0 \le h(0) \le C_0 h_0$, let $h(s)$ solve HYM. With $\widehat{K}$ as in Section 5 and $\eta = \int_{C_\tau} D_{\mathrm{tr}}(h(0), \widehat{K}(H(0)))$, one has*

$$\sup_{s \ge 0} \int_{C_\tau} d(h(s), \widehat{K}(H(s)))^2 \, dA_\tau \le \eta + C a_*. \tag{75}$$

The constant is independent of every scale ratio and of time. The conclusion permits zero component weights and a nonzero common kernel in (8). No positive lower bound on a normalized smallest eigenvalue is assumed.

Proof. The actual flow is global and has the fixed order bounds by (27). Its total derivative energy is bounded by (33), since every fibre has area four. Lemma 5.1 supplies the same bound for the comparison. Apply Lemma 4.2 to the global decomposition in Proposition 6.1. Its right-hand side is at most $\eta + C(a_* + \rho)$, uniformly for all finite time horizons. Since $\rho \le a_*$, taking the supremum over horizons proves (75). □

Uniform convergence in a family of initial metrics requires $\eta \to 0$ uniformly in that family. The theorem compares each parameter-dependent finite trajectory with its actual HYM trajectory. It asserts neither a common limiting trajectory nor a selected limiting phase in a nonzero kernel. When the active constraints are irreducible, the finite limit is o; the theorem still concerns absolute distance, not an error relative to the decaying trajectory. The next section supplies an open spatial class that enters this comparison after a negligible initial time on every effective scale.

7. Entry from spatially varying metrics

We specify an open class of initial metrics without prescribing their finite initial tuple. The tuple is selected by auxiliary flows on the fixed closed normalization components at $\alpha = 0$. Their use during a short preparation interval will be justified on the actual smoothing.

7.1. **The selected constant metric on a fixed component.** Let Y be a fixed connected closed Riemann surface of area A. For matrix functions write $\langle f \rangle = A^{-1} \int_Y f \, dA$, and let $\lambda_1 > 0$ be the first positive scalar eigenvalue. Consider the trivial holomorphic bundle in a fixed global flat frame.

Lemma 7.1. *Let $h(s)$ be HYM flow from a smooth determinant-one metric with $cI \le h(0) \le C_0 I$. There is a unique constant determinant-one matrix P such that $h(s) \to P$ uniformly. With $M(s) = \langle h(s) \rangle$, $J(s) = \log\det M(s)$ and $W(s) = \langle \|h(s) - M(s)\|^2 \rangle$, one has*

$$M'(s) \preceq 0, \qquad P = \lim_{s\to\infty} M(s), \qquad cI \le P \le C_0 I,$$

$$\frac{W}{2C_0^2} \le J \le \frac{W}{2c^2}, \qquad J(s) \le J(0) e^{-\gamma s}, \qquad \operatorname{tr}(M(s) - P) \le C_0 J(s), \qquad \gamma = \frac{2c^2}{C_0^2}\lambda_1. \tag{76}$$

For $s \ge 1$,

$$\sup_Y d(h(s), P)^2 \le C_Y (2J(0) + J(0)^2) e^{-\gamma(s-1)}. \tag{77}$$

If P, P' are selected by two initial metrics in this bounded range, then

$$A D_{\mathrm{tr}}(P, P') \le \int_Y D_{\mathrm{tr}}(h(0), h'(0)) \, dA. \tag{78}$$

Constants in these conclusions do not use derivatives of the initial metrics.

Proof. Equation (25) is $h_s = -\Delta h - Q(h)$ with $Q = 4a^{-1} h_z^* h^{-1} h_z \succeq 0$. Averaging gives $M' = -\langle Q \rangle \preceq 0$. The bounded decreasing Hermitian family $M(s)$ has a limit P with eigenvalues in $[c, C_0]$; the determinant of this limit has not yet been determined.

For $f(B) = \log\det B$ and a Hermitian matrix Z,

$$-D^2 f_B[Z, Z] = \operatorname{tr}(B^{-1} Z B^{-1} Z), \qquad C_0^{-2} \|Z\|^2 \le -D^2 f_B[Z, Z] \le c^{-2} \|Z\|^2$$

on the same matrix interval. Taylor's integral formula along the segment from M to $h(x)$, averaged over Y, has zero linear term. Since $\log\det h(x) = 0$, it gives the two bounds between J and W in (76).

Using $M^{-1} \geq C_0^{-1} I$, then $h^{-1} \geq C_0^{-1} I$, gives

$$-J' = \operatorname{tr}(M^{-1}\langle Q \rangle) \geq C_0^{-1} \langle \operatorname{tr} Q \rangle \geq C_0^{-2} \langle |dh|^2 \rangle.$$

Here $4a^{-1} \operatorname{tr}(h_z^* h_z) = |dh|^2$. This identity takes the trace before any nonconstant matrix weight is introduced; no weighted commutator has been dropped. Applying the scalar Poincaré inequality to the real matrix entries of $h - M$ gives $\langle |dh|^2 \rangle \geq \lambda_1 W \geq 2c^2 \lambda_1 J$. This proves the exponential bound for J.

It follows that $\det P = 1$. Since $M - P \succeq 0$, integration of the logarithmic determinant along the segment from P to M gives $J \geq C_0^{-1} \operatorname{tr}(M - P)$. The cross term vanishes when averaging $\|h - P\|^2$, so

$$\langle \|h - P\|^2 \rangle = W + \|M - P\|^2 \leq C_0^2 (2J + J^2).$$

This proves exponential L^2 convergence.

The constant P is an exact stationary solution. By Lemma 4.1, $D_{\mathrm{tr}}(h, P)$ is a scalar heat subsolution. The time-one scalar heat kernel on this fixed compact surface is bounded. Thus its supremum at $s \geq 1$ is at most a fixed multiple of its integral at $s - 1$. Equivalence of trace and squared matrix distances on the bounded range proves (77). Applying the integrated unforced trace comparison to two flows and passing to their uniform limits proves (78). □

The selected matrix depends on the full initial metric. For simultaneously diagonal data $h(0) = \operatorname{diag}(e^{f_1}, \dots, e^{f_r})$ with $\sum_a f_a = 0$, the scalar heat equation gives $P = \operatorname{diag}(e^{\langle f_1 \rangle}, \dots, e^{\langle f_r \rangle})$. In general neither the initial arithmetic mean nor the exponential of its mean logarithm is asserted to equal P.

7.2. **An open class on the smooth fibres.** Fix $0 < \beta < 1$. Let $\Omega_i(2\rho)$ be the closed component with the branch disks of radius 2ρ removed, identified with its region in the smooth fibre. For a smooth trace-free h_0-Hermitian logarithm S, express its restriction S_i in the harmonic component gauge and set

$$\mathcal{N}(S) = \|S\|_{C^0(C_\tau)} + \sum_i \inf_{\widetilde{S}_i|_{\Omega_i(2\rho)} = S_i} \|\widetilde{S}_i\|_{C^{6,\beta}(C_i)}. \tag{79}$$

Each infimum is over smooth Hermitian trace-free extensions to the closed component C_i. Define $\mathcal{B}_{R_0} = \{e^S : \mathcal{N}(S) < R_0\}$. For each fixed fibre this is an open nonempty class in the smooth $C^{6,\beta}$ topology. Indeed restriction to the punctured components has a bounded linear extension across their smooth boundary circles, so each infimum seminorm is continuous. No uniform estimate for this extension operator as $\rho \to 0$ is needed or claimed.

The class includes small smooth component profiles. It also permits arbitrary smooth modifications inside the removed disks and middle nodes, provided the global C^0 bound and strict norm margin are preserved. In particular their spatial derivatives there need not be uniformly bounded. Choose R_0 small enough that phases obtained from extensions with total norm below $2R_0$ lie in the finite ball of Theorem 6.2. This is possible because their order bounds imply $d_A(P, o) \leq 2\sqrt{r \sum_i A_i}\, R_0$.

Theorem 7.2. *For the fixed data of Theorem 6.2 and sufficiently small $R_0 > 0$, every initial metric in $\mathcal{B}_{R_0}$ has a selected tuple P and a common preparation time $s_* \to \infty$ such that $\sigma s_* \to 0$ and*

$$\eta_* := \int_{C_\tau} D_{\mathrm{tr}}(h(s_*), \widehat{K}(P))\, dA_\tau \longrightarrow 0. \tag{80}$$

The time and convergence bounds are uniform over this class and all permitted scale ratios. If $H_f(s)$ is the finite solution with $H_f(0) = P$, then

$$\sup_{s\geq s_*}\int_{C_\tau} d(h(s),\widehat{K}(H_f(s)))^2\,dA_\tau \leq C\{\eta_* + a_* + (\sigma s_*)^2\} \longrightarrow 0. \tag{81}$$

Different admissible extensions in (79) give tuples at distance $O(\rho)$ and reconstructions at integrated squared distance $O(\rho^2)$ at every later time.

Proof. Choose extensions $\widetilde{S}_i$ with total $C^{6,\beta}$ norm below $2R_0$. Let $\Psi_i(s)$ be HYM flow for the trivial connection on the fixed closed component, from $\exp(\widetilde{S}_i)$, and let P_i be its phase from Lemma 7.1. These auxiliary flows have $\alpha_i = 0$. Their difference from the actual component equation will be estimated explicitly.

For each integer $n \geq 1$, fixed-component parabolic theory gives a constant $C_n \geq 1$ controlling two spatial derivatives and one time derivative of all these flows on $[0, n]$. One can choose C_n nondecreasing. To justify uniformity over the profiles, the bounded initial $C^{6,\beta}$ class is precompact in $C^{4,\beta/2}$; continuous dependence for the fixed strictly parabolic equation, the order bounds and fixed-component continuation give a finite bound on each finite time interval. No assertion that C_n stays bounded as $n \to \infty$ is used.

Construct a preparation metric $K_{\mathrm{pre}}(s)$ with $\Psi_i(s)$ on $\Omega_i(2\rho)$. Use node geodesics between their values at the branch points, with parameter $-\epsilon_e \log|u_e|$, and interpolate relative logarithms on the incidence collars, including the actual unitary changes of frame. As before this is a global smooth determinant-normalized metric with fixed order bounds. On the component exterior, subtracting the trivial-connection equation from the actual D_i equation leaves integrated error at most $C_n\alpha_*$. Metric transfer contributes $C_n b^2\rho^{-N}$. The node curvature is zero and its time residual has integral at most $C_n\rho^2$.

On an incidence collar the component value differs from its value at the branch point by $O(C_n\rho)$, including the change of frame. The node logarithm differs from that point value by $O(C_n\epsilon_e|\log\rho|)$. The discrepancy derivatives of order $k \leq 2$ are therefore bounded by $C_n(\rho + \epsilon_*|\log\rho|)\rho^{-k}$. The full curvature-difference estimate used in Section 6 gives the integrated bound $C_n(\rho + \epsilon_*|\log\rho|)$, with the quadratic terms included. Consequently

$$\sup_{0\leq s\leq n}\int_{C_\tau} |\mathcal{R}(K_{\mathrm{pre}}(s))| \leq C_n\delta_*, \qquad \delta_* = \alpha_* + \epsilon_*\ell + b^2\rho^{-N} \to 0. \tag{82}$$

At time zero K_{pre} and $h(0)$ agree on every $\Omega_i(2\rho)$ and have bounded difference elsewhere. The latter region has area $O(\rho^2)$. Integrated forced trace comparison gives error at most $C\rho^2 + C_n n\delta_*$ through time n.

For sufficiently small parameters choose

$$s_* = \max\{n \in \mathbb{N} : nC_n \leq \delta_*^{-1/2}\}. \tag{83}$$

Then $s_* \to \infty$ and $\sigma s_* \to 0$, since $\sigma \leq C\delta_*$. Uniform convergence of the auxiliary flows to P_i, the component Green L^2 bounds, and the area $O(\rho^2)$ of the remaining regions show

$$\eta_* \leq C\big(\sqrt{\delta_*} + e^{-\gamma_0(s_*-1)} + \epsilon_*^2\big) \longrightarrow 0 \tag{84}$$

for a fixed $\gamma_0 > 0$ depending on the component geometries and initial order bounds. On this range trace distance and squared affine distance are equivalent, so the triangle comparison used here costs only a fixed constant.

Restart Theorem 6.2 at s_* from the finite tuple P. This first compares $h(s)$ with $\widehat{K}(H_f(s - s_*))$. Convex gradient flow on the Hadamard product is nonexpanding: first variation of

the squared distance between two solutions and convexity make its derivative nonpositive. Autonomy therefore gives

$$d_A(H_f(s-s_*),H_f(s)) \le d_A(P,H_f(s_*)) \le C\sigma s_* \qquad (s \ge s_*).$$

Lemma 5.1 transfers this difference to global integrated squared distance, proving (81).

Two choices of extensions agree on $\Omega_i(2\rho)$. Their initial component trace difference is bounded and supported on disks of total area $O(\rho^2)$. Equation (78) yields $d_A(P,P') = O(\rho)$. Nonexpansion of finite flow and (55) preserve the asserted ambiguity for all later times. □

The preparation time is negligible compared with every finite scale: $\sigma s_* \to 0$, while all finite speeds are bounded by a constant times σ. No approximation is asserted during the discarded initial layer. The tuple P depends on the spatial initial data; convergence to a single parameter-independent finite trajectory would require convergence of the selected phases and of the rescaled parameters, in addition to the estimates proved here.

8. The complete small spectrum and a three-scale family

The linearization of HYM at the parallel metric is the heat equation for the induced connection Laplacian on trace-free Hermitian endomorphisms. Write $\mathfrak{p}_r = \mathrm{Herm}_0(r)$, $d = r^2-1$, $m = 4d$, and

$$\mathcal{L}_{\tau,\alpha,U} = D^*D, \qquad \mathcal{E}_\tau(s) = \int_{C_\tau} |Ds|^2\, dA_\tau.$$

In this section the unitary transports may vary between fibres. For $X \in \mathfrak{p}_r^4$ set

$$\begin{aligned} M(X) &= \sum_i A_i \|X_i\|^2, \\ g_\alpha(X) &= \sum_{i=+,-} \alpha_i^2 \kappa_i \|[T_i, X_i]\|^2, \\ Q(X) &= g_\alpha(X) + \sum_{e=ij} w_e \|X_i - \mathrm{Ad}(U_{ij})X_j\|^2, \qquad w_e = 2\pi/L_e. \end{aligned} \tag{85}$$

These are respectively the mass form and the Hessian of (11) at o. They are defined by the component geometry, harmonic forms and node transports before solving a global analytic problem.

Theorem 8.1. *Let $\nu_0 \le \cdots \le \nu_{m-1}$ be the generalized eigenvalues of (Q,M), including zeros, and let $\lambda_0 \le \lambda_1 \le \cdots$ be those of $\mathcal{L}_{\tau,\alpha,U}$. There are $\gamma > 0$ and $e(\alpha_*, b) \to 0$ as $\alpha_*, b \to 0$ such that*

$$(1-e)\nu_k \le \lambda_k \le (1+e)\nu_k \quad (0 \le k < m), \qquad \lambda_m \ge \gamma. \tag{86}$$

Exactly m eigenvalues, counting zeros, tend to zero. The kernels of the finite and actual operators correspond exactly through (8). The estimates are uniform in all $\alpha_i^2 L_e$, all L_e/L_f, and all unitary transports, including reducible ones.

Component averaging identifies the m-dimensional spectral subspace associated with the eigenvalues tending to zero asymptotically isometrically with $(\mathfrak{p}_r^4, M)$. Let Π be the orthogonal projection onto this small-eigenvalue subspace, and let P_a be the orthogonal projection onto the image of the trial map J_a constructed below. Then

$$\|\Pi - P_a\|_{L^2 \to L^2} \le C\sqrt{\alpha_*^2 + w_*}, \qquad w_* = \max_e w_e. \tag{87}$$

On the complementary space the inverse, heat and resolvent bounds are respectively γ^{-1}, $e^{-\gamma s}$ *and* $(\gamma + \operatorname{Re} z)^{-1}$ *for* $\operatorname{Re} z \geq 0$.

The scalar graph approximation on degenerating surfaces has a substantial history, including relative estimates with arbitrary short-edge ratios [Bur90]. Here the point requiring proof is the combined component-and-node estimate, including the commutator kernel and every weaker positive scale.

8.1. **Trial sections.** Choose a small fixed radius a in the branch disks, independent of all degenerating parameters. On an elliptic component begin with the constant matrix X_i in the harmonic gauge. On $a < |u| < 2a$ interpolate it with its parallel extension having node value X_i. The primitive of the harmonic one-form obeys $|F_{ie}| \leq Ca$ and $|dF_{ie}| \leq C$. The exact bound

$$\| \operatorname{Ad}(e^{i\alpha_i T_i F_{ie}}) X_i - X_i \| \leq |\alpha_i F_{ie}| \| [T_i, X_i] \|$$

retains the component energy factor in every cutoff term. The component energy is at most $(1+Ca^2) g_\alpha(X)$. On rational components use constants. Conformal invariance transfers these energies to the fibre exactly.

On the remaining node annulus, whose conformal length is $\ell_e = L_e + 2\log a$, interpolate the two parallel-frame values affinely in logarithmic radius. Its energy is exactly $(2\pi/\ell_e)\|X_i - \operatorname{Ad}(U_e) X_j\|^2$. Denote the resulting continuous H^1 trial section by $J_a X$. Then

$$\begin{aligned} \mathcal{E}_\tau(J_a X) &\leq (1 + Ca^2) g_\alpha(X) + \sum_e \frac{2\pi}{\ell_e} \|X_i - \operatorname{Ad}(U_e) X_j\|^2, \\ 1 &\leq L_e/\ell_e \leq 1 + C_a/L_*, \\ \big| \|J_a X\|_{L^2}^2 - M(X) \big| &\leq (Ca^2 + o_a(b)) M(X). \end{aligned} \tag{88}$$

The mass estimate uses the $O(a^2)$ disk and annular areas, unitarity and bounded interpolation, and smooth metric convergence on fixed component exteriors. It is uniform in the transports and length ratios. Min–max gives the relative upper bound in (86), first choosing a small and then the parameters small. For one fixed a it also gives $\lambda_{m-1} \leq C(\alpha_*^2 + w_*)$.

8.2. **Component means with an energy-controlled frame error.** On the punctured elliptic domain $\Omega_i(a)$, let $\vartheta_{i,a}$ be the absolute harmonic representative of the restriction class of ϑ_i, with zero normal component at the boundary. Write $\vartheta_i = \vartheta_{i,a} + df_{i,a}$ and $\kappa_{i,a} = \|\vartheta_{i,a}\|_{L^2}^2$. Then $\kappa_{i,a} \to \kappa_i > 0$. For completeness, the restriction of ϑ_i is a competitor for the energy minimization, giving a uniform upper bound. A weakly convergent harmonic subsequence on compact subsets has finite L^2 energy across every puncture. Its holomorphic representation has no negative Laurent terms with finite L^2 norm, so it extends across the puncture. Its periods, including zero puncture periods, are those of ϑ_i. Uniqueness of the harmonic representative on the closed component and lower semicontinuity prove the asserted norm convergence.

Gauge by $e^{i\alpha_i T_i f_{i,a}}$ and diagonalize T_i. Each matrix entry then has connection $d - i\beta\vartheta_{i,a}$ with $\beta = \alpha_i(t_b - t_c)$. Decompose an entry as $v = z + w$, where w has mean zero. Coclosedness and the zero normal condition give $\langle dw, \vartheta_{i,a} z \rangle = 0$. The fixed-domain Poincaré inequality and a small-β perturbation give $\|w\| \leq C_a \|(d - i\beta\vartheta_{i,a}) w\|$. The remaining cross term is bounded by $C_a \beta^2 |z| \|w\|$. Completing the square therefore yields

$$\|(d - i\beta\vartheta_{i,a}) v\|^2 \geq \frac{1}{2} \|(d - i\beta\vartheta_{i,a}) w\|^2 + (\beta^2 \kappa_{i,a} - C_a \beta^4) |z|^2. \tag{89}$$

Summing matrix entries defines a real linear average $s \mapsto X_i(s)$ and gives

$$\begin{aligned} \mathcal{E}_i(s) &\geq (1 - C_a\alpha_*^2)\alpha_i^2\kappa_{i,a}\|[T_i, X_i]\|^2, \\ \|w_i\|_{H^1}^2 + \alpha_i^2\|[T_i, X_i]\|^2 &\leq C_a\mathcal{E}_i(s). \end{aligned} \tag{90}$$

No division by α_i or an eigenvalue difference is used. On rational components take the ordinary mean. These estimates concern actual component energy by conformal invariance.

Let c_{ie} be the boundary-circle mean in the parallel node frame. The change from the averaging gauge to this frame acting on X_i is at most $C_a|\alpha_i|\|[T_i, X_i]\|$. The remaining error is bounded by the H^1 trace norm of w_i. Thus

$$\|c_{ie} - X_i(s)\| \leq C_a\mathcal{E}_i(s)^{1/2}. \tag{91}$$

Replacing this by an error proportional to $|\alpha_i|\|X_i\|$ would lose relative control near a changing commutant.

8.3. **The combined lower form.** Partition the energy into the disjoint truncated components and their complementary node annuli: $\mathcal{E}_\tau = \mathcal{E}_{\mathrm{comp}} + \mathcal{E}_{\mathrm{neck}}$. Angular averaging and the one-dimensional Dirichlet principle give

$$\sum_e \frac{2\pi}{\ell_e}\|c_{ie} - \mathrm{Ad}(U_e)c_{je}\|^2 \leq \mathcal{E}_{\mathrm{neck}}.$$

Apply the triangle inequality in the direct sum of edge spaces weighted by $\sqrt{w_e}$. Using $w_e \leq 2\pi/\ell_e$ and (91) gives

$$\left(\sum_e w_e\|X_i - \mathrm{Ad}(U_e)X_j\|^2\right)^{1/2} \leq \mathcal{E}_{\mathrm{neck}}^{1/2} + C_a\sqrt{w_*}\,\mathcal{E}_{\mathrm{comp}}^{1/2}. \tag{92}$$

With $\delta = C_a\sqrt{w_*}$ use $(x + \delta y)^2 \leq (1 + \delta)x^2 + (\delta + \delta^2)y^2$. Meanwhile (90) and $\kappa_{i,a} \to \kappa_i$ give $g_\alpha(X) \leq (1 + \theta(a) + C_a\alpha_*^2)\mathcal{E}_{\mathrm{comp}}$, where $\theta(a) \to 0$. Adding the two estimates yields

$$Q(X(s)) \leq \{1 + \theta(a) + C_a\alpha_*^2 + O_a(\sqrt{w_*})\}\mathcal{E}_\tau(s). \tag{93}$$

The component and annular energies have not been counted twice. This estimate is relative to their actual sum and contains no smallest scale in a denominator.

8.4. **Mass and completeness.** We give the mass estimate needed both for the lower eigenvalue bound and for completeness of the mode list. Let s be a unit eigensection with eigenvalue $\lambda \leq 1$. Equation (90) bounds its mean-zero component parts in H^1 by $C_a\sqrt{\lambda}$. On fixed collars, interior elliptic estimates improve the boundary error to the same order in supremum norm. In the equation for the mean-zero part, the Laplacian of the gauged constant is $\alpha_i^2|\vartheta_{i,a}|^2\,\mathrm{ad}(T_i)^2X_i$. Its norm is bounded by $C_a|\alpha_i|\sqrt{\mathcal{E}_i}$ using (90). The other inhomogeneous term is λs. Enlarge the fixed collar across the circle before applying the interior estimate. In the actual conformal metric the connection Laplacian is multiplied by a bounded smooth scalar there, without a new first-order term. The leading boundary values have size at most $C\|X_i\|$ with C independent of a, because the changes of frame are unitary.

On each node annulus split the section into its entrywise harmonic Dirichlet extension h and a zero-boundary remainder v. The maximum principle and annular area $O(a^2)$ bound the mass of h by $Ca^2 + C_a\sqrt{\lambda}$. Orthogonality of Dirichlet energies gives $\mathcal{E}(v) \leq \mathcal{E}(s)$. The zero-boundary Poincaré estimate

$$\|v\|_{L^2(dA_\tau)}^2 \leq Ca^2\mathcal{E}(v) \tag{94}$$

is uniform in the annular modulus. To verify it, the metric is bounded by a fixed multiple of $|du|^2 + |dv_{\mathrm{branch}}|^2$, with $v_{\mathrm{branch}} = \tau_e/u$. For the first branch, the zero outer boundary value and radial Cauchy–Schwarz give $|v(r,\theta)|^2 \le \log(a/r)\int_r^a |\partial_t v(t,\theta)|^2 t\,dt$. Integration against $r\,dr\,d\theta$ is bounded by Ca^2 times the radial energy. The other branch gives the same estimate for its area term, using the zero boundary value at its outer circle. This proves (94).

The component mean decomposition, metric convergence on fixed exteriors, and these annular bounds now imply that on any span of at most $m+1$ orthonormal eigensections with eigenvalues at most $\Lambda \le 1$,

$$|M(X(s)) - \|s\|_{L^2}^2| \le \{Ca^2 + C_a\sqrt{\Lambda} + o_a(b)\}\|s\|_{L^2}^2. \tag{95}$$

Passing from individual eigensections to this span costs only its fixed dimension. The leading omitted-area constant is independent of a.

If λ_m tended to zero along a joint degeneration, first choose a small and then take the parameters far along that sequence. Equation (95) would make averaging injective on an $(m+1)$-dimensional space into $\mathfrak{p}_r^4$, a contradiction. This proves the uniform complementary gap. The trial upper bound already puts m eigenvalues below it.

For the first $k+1$ eigensections use $\Lambda = C(\alpha_*^2 + w_*)$ in (95). Their averages form a $(k+1)$-dimensional space. Finite min–max gives a section in this span with $Q(X(s)) \ge \nu_k M(X(s))$. Combining this with (93) proves the relative lower bound. For any requested error first choose a, then choose α_*, b independently small. These quantifiers give the function $e(\alpha_*, b) \to 0$ in the theorem, uniformly in all remaining ratios and transports.

The zero set of Q is exactly (8), since every node weight is positive. Section 2 identifies these tuples with all global parallel endomorphisms, which are precisely the kernel of D^*D. Thus zeros agree exactly through all kernel changes.

Choose $a = a(\alpha_*, b)$ tending sufficiently slowly to zero that the preceding errors vanish and $|\log a|/L_* \to 0$. The averaging and trial mass estimates are then asymptotic isometries. On the complement of the small-eigenvalue subspace, the gap gives

$$\|(1-\Pi)J_a X\|^2 \le \gamma^{-1}\mathcal{E}_\tau(J_a X) \le C(\alpha_*^2 + w_*)M(X).$$

The spaces have equal dimension and the trial mass has a uniform lower bound. Equality of their maximal principal angle with the norm of the difference of orthogonal projections proves (87). The spectral theorem gives the stated inverse, heat and resolvent estimates on the complement. On the positive small-eigenvalue subspace the smallest rate and inverse norm instead have the relative comparison with the smallest positive ν_k; they need not stay bounded away from zero and infinity. This completes the proof of Theorem 8.1.

8.5. **A three-scale family.** Order the vertices as E_+, R_p, E_-, R_q, numbered $0,1,2,3$. Use identity transports on 0–1–2–3 and $U = (I + i\sigma_x)/\sqrt{2}$ on 3–0. Set $T_+ = \sigma_z$, $\alpha_- = 0$, and, for $h > 0$, take

$$w_{01} = w_{12} = w_{23} = h, \quad w_{30} = h^2, \quad \alpha_+ = h^2/\sqrt{\kappa_+}, \quad \tau_e = \exp(-2\pi/w_e). \tag{96}$$

These are genuine parameters of the compact family. Section 2 proves stability for all sufficiently small $h > 0$.

In the Hilbert–Schmidt orthonormal Pauli basis $(\sigma_x, \sigma_y, \sigma_z)/\sqrt{2}$ at each vertex, $M = I$ and the matrix of Q is

$$Q_h = h(L_{P_4} \otimes I_3) + h^2 D_{30}^T D_{30} + h^4\,\mathrm{diag}(4,4,0,0,\dots,0), \tag{97}$$

where

$$L_{P_4} = \begin{pmatrix} 1 & -1 & 0 & 0 \\ -1 & 2 & -1 & 0 \\ 0 & -1 & 2 & -1 \\ 0 & 0 & -1 & 1 \end{pmatrix}, \qquad D_{30}X = X_3 - RX_0, \qquad R = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & -1 & 0 \end{pmatrix}.$$

The positive path eigenvalues are $2-\sqrt{2}, 2, 2+\sqrt{2}$. Thus nine finite eigenvalues are these three coefficients times h, each with multiplicity three, up to $O(h^2)$. The isometric constant-tuple inclusion $Z \mapsto (Z, Z, Z, Z)/2$ compresses the cycle term to $\operatorname{diag}(0, \frac{1}{2}, \frac{1}{2})$ and the component term to $\operatorname{diag}(1, 1, 0)$. Schur complementation across the order-h complement therefore gives two eigenvalues $\frac{1}{2}h^2 + O(h^3)$.

For the last eigenvalue one must use its invariant sector, rather than this absolute remainder. The σ_x sector is invariant, its graph operator has exact constant kernel, and its positive gap is at least $(2-\sqrt{2})h$. The order-h^4 perturbation has expectation h^4 on its unit constant vector. The complementary inverse bound gives the last eigenvalue $h^4 + O(h^7)$. As a check on the displayed finite matrix, its determinant is

$$\det Q_h = 16h^{17}(1+3h)(1+2h^2+6h^3). \tag{98}$$

All entries and the compressions used in this calculation are printed above.

Corollary 8.2. *On the stable compact family (96), nine actual eigenvalues have leading coefficients $2-\sqrt{2}, 2, 2+\sqrt{2}$, each three times, at order h; two are asymptotic to $\frac{1}{2}h^2$; one is asymptotic to h^4. The next eigenvalue has a uniform positive lower bound. On the fixed small basin of Theorem 6.2, actual nonlinear HYM trajectories are approximated by (12) through time h^{-4} and all later times, with absolute integrated squared-distance error at most $\eta + O(h(1+|\log h|)^2)$. The spatial initial class of Theorem 7.2 has the corresponding all-time conclusion after preparation.*

Proof. Apply Theorem 8.1 to the finite eigenvalue estimates. Apply Theorems 6.2 and 7.2 separately to the same fixed matrices and transports. Here $\epsilon_* = h/(2\pi)$, $\alpha_* = h^2/\sqrt{\kappa_+}$, and $b = e^{-2\pi/h}$, so $a_* = O(h(1+|\log h|)^2)$. The nonlinear conclusion follows from the residual and energy proof, not from the eigenvalue asymptotics. □

9. Further questions

The spectral comparison is uniform when the transports change, whereas the all-time force estimate fixes their constraint spaces. The counterexample in Section 3 identifies the failure of one proof estimate, not a failure of HYM tracking. An extension through varying transports would require a stronger estimate for the residual or a comparison that avoids the separate force integral. The energy comparison itself has no such fixed-transport restriction.

The small finite basin is also a restriction of the force argument. Convexity confines trajectories from arbitrary bounded finite balls, and the exact HYM flow has global order bounds. Neither fact by itself supplies the uniform separate-force estimate on those larger balls. A global version of that estimate, or another integrability mechanism, would enlarge the spatial initial class.

Our bundle family is flat unitary and therefore polystable on every smooth fibre. The stable-to-strictly-polystable transition is genuine, but the construction contains no unstable or nonunitary stratum. It has a locally free bundle on the smooth total space and no concentration of curvature in the parallel reference metrics. Singular graded quotients and bubbling would introduce local concentration data not present in this model.

Finally, the all-time error is absolute. At reducible parameters the finite trajectory can tend to a nonidentity parallel tuple, and its phase is selected by the initial metric. A uniform infinite-time limit over degenerating parameters is a different question from uniform approximation of these individual trajectories. The present theorem concerns metrics on bundle fibres. Extending it to distances on the moduli space of bundles, or to a higher-dimensional degenerating base, requires additional geometric and analytic arguments.

References

[Bur90] M. Burger, *Small eigenvalues of Riemann surfaces and graphs*, Math. Z. **205** (1990), 395–420. doi:10.1007/BF02571252.

[DW00] G. D. Daskalopoulos and R. A. Wentworth, *The Yang–Mills flow near the boundary of Teichmüller space*, Math. Ann. **318** (2000), no. 1, 1–42. doi:10.1007/s002080000098.

[Don85] S. K. Donaldson, *Anti self-dual Yang–Mills connections over complex algebraic surfaces and stable vector bundles*, Proc. London Math. Soc. (3) **50** (1985), no. 1, 1–26. doi:10.1112/plms/s3-50.1.1.

[HKKP18] F. Haiden, L. Katzarkov, M. Kontsevich and P. Pandit, *Iterated logarithms and gradient flows*, preprint, 2018. arXiv:1802.04123v1.

[HK14] J. H. Hubbard and S. Koch, *An analytic construction of the Deligne–Mumford compactification of the moduli space of curves*, J. Differential Geom. **98** (2014), no. 2, 261–313. doi:10.4310/jdg/1406552251. arXiv:1301.0062v1.

[JW92] L. Ji and R. Wentworth, *Spectral convergence on degenerating surfaces*, Duke Math. J. **66** (1992), no. 3, 469–501. doi:10.1215/S0012-7094-92-06615-4.

[JW97] L. Ji and R. Wentworth, *Correction to "Spectral convergence on degenerating surfaces"*, Duke Math. J. **90** (1997), no. 1, 205–207. doi:10.1215/S0012-7094-97-09007-4.

[Kob87] S. Kobayashi, *Differential Geometry of Complex Vector Bundles*, Publ. Math. Soc. Japan **15**, Kanô Memorial Lectures **5**, Iwanami Shoten and Princeton University Press, 1987. Authorized 2013 retypesetting.

[Lam26] S. T. Lam, *Semistability and asymptotics of geometric flows*, preprint, 2026. arXiv:2606.09235v1.

[NS65] M. S. Narasimhan and C. S. Seshadri, *Stable and unitary vector bundles on a compact Riemann surface*, Ann. of Math. (2) **82** (1965), no. 3, 540–567. doi:10.2307/1970710.

[Oht20] S. Ohta, *Self-contracted curves in CAT(0)-spaces and their rectifiability*, J. Geom. Anal. **30** (2020), 936–967. doi:10.1007/s12220-018-00126-7. Theorem and equation numbering follows arXiv:1711.09284v2.

[Wue26a] B. J. Wuebben, *Effective Hermitian–Yang–Mills flow on nodal degenerations of curves*, preprint, 10 November 2026. doi:10.5281/zenodo.24120966.

[Wue26b] B. J. Wuebben, *Hermitian–Yang–Mills flow and gluing on degenerating curves*, preprint, 11 November 2026. doi:10.5281/zenodo.24149197.